4/23/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\GEN\002-C101.dwg

GRAND CANYON LODGE PARKING

GUEST CABINS

BALLFIELD SITE

CONCESSIONS AREA

EXISTING EMERGENCY MEDICAL SERVICES FACILITY

NORTH RIM WASTEWATER TREATMENT PLANT

STATE ROUTE 67

TP453
TP477
TP452
TP451
V8PH24AZS
V8PH23AZS
TP450
TP459
TP460
TP476
TP474
TP475
TP472
TP470
TP471
TP469
TP473
TP466
TP467
TP465
TP458
TP457
TP456
TP455
TP461
TP462
TP463
TP464
TP468
TP474
TP100
TP101
TP102
TP103
V1P101AZS
V8PH09AZS
V8P091AZS
V1PH08AZS
V1P081AZS
TP430
TP418
TP419
V6PH15AZS
V6PH14AZS
V6PH13AZS
V6PH12AZS
V6P111AZS
V6PH11AZS
V6PH09AZS
V1PH06AZS
V1PH07AZS
V1P051AZS
V1PH05AZS
V1PH03AZS
V1PH04AZS
V1P041AZS
V1PH02AZS
V8PH28AZS
TP429
V1PH11AZS
V1PH10AZS
V8PH16AZS
V8PH03AZS
V6PH02AZS
V6PH01AZS
V6PH05AZS
V6PH08AZS
V8PH07AZS
V1PH17AZS
V6PH10AZS
TP2000
TP417
TP416
TP415
V1PH15AZS
V1PH16AZS
V8PH12AZS
V8PH13AZS
V8PH10AZS
V8PH11AZS
TP431
TP426
TP428
TP432
TP433
TP427
TP423
TP412
TP413
TP411
TP409
TP410
TP408
TP404
TP402
TP424
TP425
TP422
TP420
TP421
V8PH07AZS
TP407
TP406
TP405
TP403
TP408
TP401
V7PH07AZS
V7PH06AZS
V7PH04AZS
V7PH05AZS
V7PH09AZS
V7PH10AZS
V7P091AZS
V7P092AZS
V7PH08AZS
V7P081AZS
V7PH12AZS
V7PH02AZS
TP112
TP106
TP111
TP109
TP105
TP108
TP104
TP107
TP113
TP110

GENERAL NOTES:

A. SEE SHEET 002C102 FOR POTHOLE POINT TABLE.

300 0 300 600
SCALE OF FEET
1" = 300'

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
United States of America
ARIZONA, U.S.A.

A/E FIRM	
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	
DESIGNED:	G. YOUNG
DRAWN:	P. OXBORROW
TECH. REVIEW:	M. GOUGH/R. RHODES
DATE:	MAY 2026
SUB SHEET NO.	002 C101
TITLE OF SHEET	GENERAL SURVEY POTHOLE PLAN (NORTH RIM)
	NORTH RIM - GRAND CANYON NATIONAL PARK
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	18 OF 99

NORTH RIM AREA POTHOLE POINT TABLE				
POINT NAME	NORTHING	EASTING	ELEVATION	DESCRIPTION
TP100	13146091.43	1329028.9500	8241.42	NA 60D
TP101	13146199.34	1329086.6310	8232.15	NA 60D
TP102	13145811.40	1328894.5990	8238.60	NA 60D
TP103	13145603.95	1328770.4710	8231.50	NA 60D
TP104	13145709.06	1328608.3830	8228.16	NA 60D
TP105	13145837.42	1328619.1770	8231.65	NA 60D
TP106	13146071.87	1328696.3800	8241.99	NA 60D
TP107	13145744.55	1328573.6880	8226.30	NA 60D
TP108	13145954.64	1328436.8700	8225.61	NA 60D
TP109	13146089.55	1328566.1010	8237.17	NA 60D
TP110	13145893.86	1328465.0590	8222.94	NA 60D
TP111	13146221.19	1328550.1790	8239.99	NA 60D
TP112	13146330.51	1328735.4690	8250.61	NA 60D
TP113	13145914.58	1328515.8440	8228.64	NA 60D
TP400	13151333.55	1326579.7120	8338.82	NA 60D
TP401	13151246.99	1326522.3980	8335.19	NA MAG
TP402	13151256.01	1326734.2430	8341.53	NA 60D
TP403	13151193.59	1326650.8500	8335.90	NA MAG
TP404	13151224.24	1326828.4860	8343.84	NA 60D
TP405	13151149.98	1326774.4940	8337.55	NA MAG
TP406	13151107.13	1326836.8550	8337.87	NA MAG
TP407	13151040.58	1326930.3200	8337.60	NA MAG
TP408	13151233.26	1327002.2250	8349.15	NA 60D
TP409	13151194.12	1327123.5430	8351.59	NA 60D
TP410	13151138.63	1327017.7150	8346.75	NA 60D
TP411	13151255.04	1327207.0340	8355.08	NA 60D
TP412	13151161.83	1327296.3930	8355.30	NA 60D
TP413	13151063.40	1327254.3720	8352.34	NA 60D
TP415	13150843.94	1327370.8040	8351.96	NA 60D
TP416	13150787.06	1327485.6380	8351.09	NA 60D
TP417	13150762.83	1327592.6860	8350.32	NA 60D
TP418	13148944.24	1329026.4510	8321.12	NA 60D
TP419	13148929.26	1328907.3520	8317.98	NA 60D
TP420	13150793.45	1327082.8210	8339.28	NA 60D
TP421	13150644.55	1326925.3710	8331.09	NA MAG
TP422	13150386.22	1327105.9410	8333.39	NA 60D
TP423	13150610.74	1327304.2170	8345.21	NA 60D
TP424	13150203.16	1327205.6780	8331.09	NA 60D
TP425	13150149.97	1327067.1180	8321.70	NA MAG
TP426	13150190.21	1327543.9270	8325.40	NA 60D
TP427	13150013.69	1327346.4260	8310.48	NA 60D
TP428	13150296.46	1327430.8980	8328.64	NA 60D
TP429	13148850.40	1328916.6520	8316.12	NA 60D
TP430	13148889.92	1328990.8130	8321.17	NA MAG
TP431	13150428.88	1327501.7080	8335.18	NA 60D

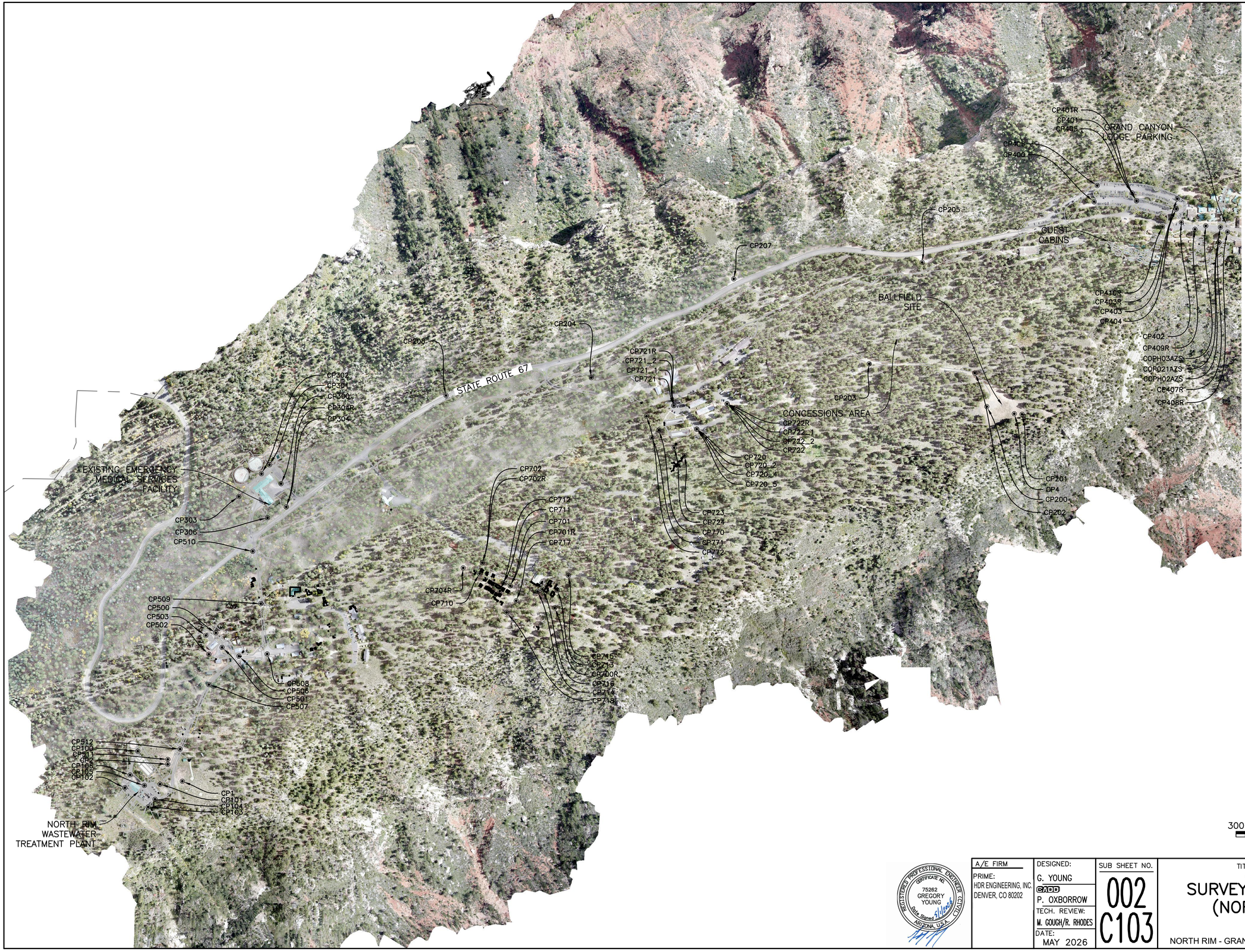
NORTH RIM AREA POTHOLE POINT TABLE				
POINT NAME	NORTHING	EASTING	ELEVATION	DESCRIPTION
TP432	13150330.90	1327356.0520	8334.02	NA MAG
TP433	13150135.58	1327390.2620	8319.75	NA SPIKE
TP450	13144374.50	1329774.2020	8217.45	NA 60D
TP451	13144361.98	1330046.5780	8231.23	NA 60D
TP452	13144327.19	1330150.5060	8231.91	NA 60D
TP453	13144442.47	1329729.4550	8216.65	NA 60D
TP454	13144538.28	1329704.6720	8217.87	NA 60D
TP455	13144432.26	1329658.7030	8206.04	NA 60D
TP456	13144391.80	1329672.7080	8204.82	NA 60D
TP457	13144351.45	1329706.3740	8207.00	NA 60D
TP458	13144312.51	1329728.9590	8208.63	NA 60D
TP459	13144319.73	1329755.8460	8214.32	NA 60D
TP460	13144501.23	1329787.2470	8224.43	NA 60D
TP461	13144513.42	1329643.2750	8207.70	NA 60D
TP462	13144581.50	1329645.7750	8209.14	NA 60D
TP463	13144641.02	1329638.2200	8204.92	NA 60D
TP464	13144699.61	1329643.3590	8204.29	NA 60D
TP465	13144789.37	1329638.9020	8202.28	NA 60D
TP466	13144912.06	1329655.1510	8203.10	NA 60D
TP467	13144859.02	1329703.7260	8215.95	NA 60D
TP468	13144813.07	1329778.9740	8226.10	NA 60D
TP469	13145003.34	1329672.9720	8199.26	NA 60D
TP470	13145085.82	1329718.1750	8202.93	NA 60D
TP471	13145050.31	1329760.4740	8216.76	NA 60D
TP472	13145134.87	1329780.3440	8212.52	NA 60D
TP473	13144958.48	1329736.2140	8217.62	NA 60D
TP474	13144836.13	1329737.1020	8221.05	NA 60D
TP475	13144793.38	1329820.9550	8229.96	NA 60D
TP476	13144523.41	1329749.7650	8222.16	NA 60D
TP477	13144515.70	1329822.2960	8227.05	NA 60D
TP2000	13150657.42	1327725.8160	8349.00	NA 60D
V1P041AZS	13147778.50	1329127.9120	8294.16	GTB
V1P051AZS	13147741.47	1329146.9290	8292.44	GTB
V1P081AZS	13148687.17	1328923.8010	8313.27	GTB
V1P101AZS	13148769.34	1328909.1920	8314.01	GTB
V1PH02AZS	13146996.35	1329098.1980	8272.35	GTB
V1PH03AZS	13147762.74	1329138.3180	8293.27	GTB DRY
V1PH04AZS	13147778.84	1329127.3440	8294.17	GTB
V1PH05AZS	13147741.22	1329148.6150	8292.64	GTB
V1PH06AZS	13148243.64	1329010.6370	8306.24	GTB
V1PH07AZS	13148280.85	1328974.6510	8306.95	GTB
V1PH08AZS	13148686.93	1328923.0000	8313.27	GTB
V1PH10AZS	13148769.28	1328908.2360	8313.86	GTB
V1PH11AZS	13148834.32	1328896.3080	8314.14	GTB
V1PH15AZS	13150013.71	1328294.9670	8346.23	GTB

NORTH RIM AREA POTHOLE POINT TABLE				
POINT NAME	NORTHING	EASTING	ELEVATION	DESCRIPTION
V1PH16AZS	13150206.80	1328131.2500	8348.78	GTB
V1PH17AZS	13150551.26	1327993.5770	8354.57	GTB
V6P111AZS	13150502.97	1328301.6110	8359.34	GTB
V6PH02AZS	13150709.26	1328376.3640	8360.49	GTB
V6PH03AZS	13150690.72	1328367.8800	8352.18	GTB
V6PH04AZS	13150676.37	1328331.1020	8359.62	GTB
V6PH05AZS	13150664.05	1328321.8340	8359.08	GTB
V6PH07AZS	13150599.83	1328211.2000	8358.46	GTB
V6PH08AZS	13150605.94	1328221.9830	8358.54	GTB
V6PH09AZS	13150550.78	1328117.7710	8356.69	GTB
V6PH10AZS	13150598.02	1327889.9040	8352.13	GTB
V6PH11AZS	13150501.45	1328300.1330	8359.31	GTB
V6PH12AZS	13150580.80	1328343.7870	8359.80	GTB
V6PH13AZS	13150566.65	1328357.0360	8359.94	GTB
V6PH14AZS	13150594.68	1328392.5270	8360.34	GTB
V6PH15AZS	13150585.02	1328392.7950	8360.45	GTB
V6PH16AZS	13150610.04	1328415.4550	8360.44	GTB
V7P081AZS	13151406.70	1326215.2420	8323.02	GTB
V7P091AZS	13151349.89	1326323.3070	8329.33	GTB
V7P092AZS	13151349.47	1326323.0350	8329.30	GTB
V7PH02AZS	13151506.82	1326048.1540	8301.88	GTB DRY
V7PH04AZS	13151653.67	1326156.9140	8302.18	GTB
V7PH05AZS	13151656.03	1326165.0310	8302.26	GTB
V7PH06AZS	13151544.83	1326192.5270	8302.83	GTB
V7PH07AZS	13151498.42	1326242.5220	8322.89	GTB
V7PH08AZS	13151405.47	1326215.1560	8323.10	GTB
V7PH09AZS	13151350.15	1326323.9360	8329.29	GTB
V7PH10AZS	13151363.30	1326328.6130	8329.13	GTB
V7PH12AZS	13151446.60	1326095.4800	8303.86	GTB
V8P091AZS	13148726.46	1328916.6590	8313.68	GTB
V8PH07AZS	13150672.29	1326918.4120	8331.00	GTB
V8PH09AZS	13148726.24	1328915.9440	8313.67	GTB
V8PH10AZS	13148811.96	1327552.6870	8299.60	GTB
V8PH11AZS	13149512.16	1327641.8810	8317.60	GTB DRY
V8PH12AZS	13148276.13	1327782.0030	8287.86	GTB
V8PH13AZS	13148635.63	1327995.7910	8296.39	GTB DRY
V8PH16AZS	13147910.69	1328625.8680	8290.97	GTB
V8PH23AZS	13144345.99	1329827.1130	8221.29	GTB DRY
V8PH24AZS	13144324.09	1329869.7530	8222.84	GTB
V8PH28AZS	13146839.28	1329062.4470	8269.75	GTB

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO.
75262
GREGORY YOUNG
ARIZONA, U.S.A.
Date Signed 5/11/26

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: M. GOUGH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 002 C102	TITLE OF SHEET GENERAL SURVEY POTHOLE TABLE (NORTH RIM) NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 19 OF 99	

4/23/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\GEN\002-C101.dwg



GENERAL NOTES:

A. SEE SHEET 002C104 FOR SURVEY CONTROL TABLE.

300 0 300 600
SCALE OF FEET 1" = 300'

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75282
GREGORY YOUNG
Date Signed
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: M. COUGH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 002 C103	TITLE OF SHEET GENERAL SURVEY CONTROL (NORTH RIM) NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 20 OF 99

4/23/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\GEN\002-C101.dwg

NORTH RIM AREA CONTROL POINT TABLE				
POINT NAME	NORTHING	EASTING	ELEVATION	DESCRIPTION
COP021AZS	13144381.78	1329801.7960	8220.09	GTB
COPH02AZS	13144380.74	1329800.4570	8220.00	GTB
COPH03AZS	13144467.41	1329911.4380	8229.44	GTB
CP1	13151283.39	1326213.3970	8323.90	CRP Set
CP2	13151380.85	1326326.3620	0.00	AZS HDR GRCA North Rim mapping
CP4	13145743.15	1328660.6850	0.00	AZS HDR GRCA North Rim mapping
CP100	13151580.36	1326413.7500	8327.31	CP AZS CRP
CP101	13151432.11	1326194.0070	8320.71	CP AZS CRP
CP102	13151665.92	1326169.9930	8302.23	NA MAG AZS CP
CP103	13151511.02	1326042.6660	8301.74	CP AZS CRP
CP104	13151465.40	1326093.4000	8302.92	NA MAG AZS CP
CP105	13151536.46	1326184.9690	8302.97	NA MAG AZS CP
CP106	13151631.25	1326255.4820	8305.75	CP AZS CRP
CP200	13145925.99	1328708.0576	8237.88	
CP201	13145691.72	1328657.2780	8231.43	CRP Set
CP202	13146005.96	1328931.3110	8244.32	CRP AZS CRP
CP203	13146709.83	1328992.9360	8265.48	CP AZS CRP
CP204	13148560.20	1328900.3940	8310.62	CP AZS CRP
CP205	13146356.81	1329668.0840	8221.02	CP AZS CRP
CP207	13147611.76	1329558.0770	8235.62	CP AZS CRP
CP208	13149527.23	1328781.8590	8338.40	NA MAG
CP300	13150626.13	1328194.4450	8359.18	CRP Set
CP301	13150703.04	1328320.8020	8361.50	NA 60 IN STUMP
CP302	13150756.48	1328261.0380	8359.27	CP AZS CRP
CP303	13150907.64	1328109.2530	8356.48	CP AZS CRP
CP304	13150591.28	1328040.0120	8356.21	CP COPPER MON IN CONC CP COPPER
CP304R	13150591.26	1328040.0470	8356.23	CP COPPER MON
CP306	13150715.84	1327967.4080	8355.36	CP AZS CRP
CP400	13145211.26	1330072.6140	8251.59	NA MAG
CP401	13144928.96	1330077.5200	8246.53	NA MAG

NORTH RIM AREA CONTROL POINT TABLE				
POINT NAME	NORTHING	EASTING	ELEVATION	DESCRIPTION
CP401R	13144928.99	1330077.5590	8246.34	NA MAG
CP402	13144630.17	1329935.6690	8235.15	CRP Set
CP403	13144663.91	1330023.3520	8239.58	CP NPS AP861 CRP CP NPS AP861 C
CP403R	13144663.92	1330023.3650	8239.47	CP NPS BC AP861
CP404	13144657.69	1330091.3100	8240.81	NA MAG
CP405	13144958.09	1330131.0090	8247.74	NA MAG
CP406	13145190.74	1330179.5110	8253.46	NA MAG
CP407R	13144370.14	1329902.4280	8225.39	NA MAG SPIKE
CP408R	13144323.08	1329869.6250	8222.86	NA MAG
CP409R	13144539.60	1329888.6130	8232.17	NA MAG SPIKE
CP410R	13144755.37	1329905.2990	8237.31	NA MAG SPIKE
CP500	13151060.01	1327201.3590	8351.74	NA MAG IN STUMP
CP501	13151018.86	1327104.4430	8347.78	NA MAG
CP502	13151106.30	1327058.9220	8348.36	NA MAG
CP503	13151126.87	1327184.3930	8350.30	NA MAG
CP506	13150906.83	1327046.2250	8339.63	NA MAG IN ASPH
CP507	13151106.68	1326857.0710	8338.11	NA MAG IN ASPH
CP508	13150718.65	1327061.1570	8337.74	NA MAG IN ASPH
CP509	13150761.69	1327395.8030	8350.38	NA MAG IN ASPH
CP510	13150814.37	1327744.3980	8351.71	NA MAG IN ASPH
CP511	13151380.07	1326358.9450	8331.05	CP AZS CRP
CP512	13151298.14	1326427.4910	8333.70	NA MAG IN ASPH
CP700R	13148803.80	1327367.5240	8294.41	CRP AZS
CP701	13149099.69	1327514.6450	8307.51	CRP AZS
CP701R	13149099.74	1327514.6650	8307.51	CRP AZS
CP702	13149285.51	1327668.2840	8316.24	CRP AZS
CP702R	13149285.62	1327668.2620	8316.05	CRP
CP704R	13149412.68	1327634.3200	8317.52	CRP SET
CP710	13149303.42	1327596.3220	8314.19	NA
CP711	13149142.64	1327535.0850	8308.97	NA 10”

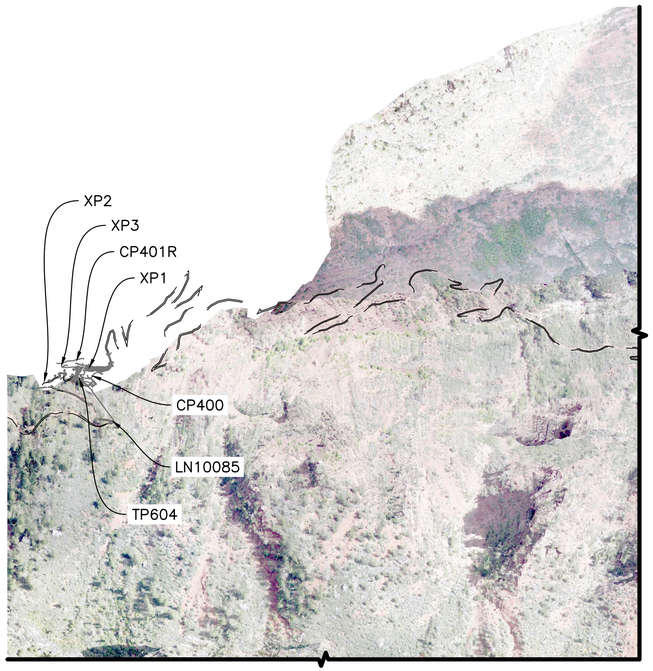
NORTH RIM AREA CONTROL POINT TABLE				
POINT NAME	NORTHING	EASTING	ELEVATION	DESCRIPTION
CP712	13149169.74	1327479.6820	8309.20	NA
CP713	13148785.65	1327484.1070	8295.36	NA
CP714	13148916.81	1327477.2300	8301.43	NA 60D
CP715	13148712.09	1327582.7370	8297.27	NA
CP716	13148885.72	1327492.3100	8300.94	NA 60D
CP717	13149060.21	1327415.3720	8304.61	NA
CP718	13149124.35	1327336.9460	8304.32	NA
CP720	13147857.58	1328557.7740	8287.05	CRP AZS
CP720_2	13147857.52	1328557.8330	8287.12	CRP AZS
CP720_4	13147857.60	1328557.8240	8287.06	CRP AZS
CP720_5	13147857.64	1328557.8040	8287.11	CRP AZS
CP721	13148009.88	1328715.2570	8297.15	CRP AZS
CP721_1	13148009.82	1328715.2530	8297.09	CRP AZS
CP721_2	13148009.83	1328715.2750	8297.14	CRP AZS
CP721R	13148009.84	1328715.2710	8297.12	NA MAG
CP722	13147655.69	1328726.3880	8292.95	NA 60D
CP722_1	13147655.66	1328726.3900	8292.93	NA 60D
CP722_2	13147655.66	1328726.3790	8292.94	NA 60D
CP722R	13147655.74	1328726.4860	8292.90	NA 60D
CP723	13147938.34	1328381.9540	8266.30	NA 60D
CP724	13148008.13	1328434.2070	8270.71	NA 60D
CP770	13148110.48	1328550.9540	8283.98	NA 60D
CP771	13148156.99	1328501.1070	8278.14	NA 60D
CP772	13148192.40	1328643.3800	8293.18	NA 60D

ABBREVIATIONS	
ASPH	ASPHALT
AZS	ARIZONA SURVEYOR
BC	BACK OF CURB
CP	CONTROL POINT
CRP	COMMON/CENTRAL REFERENCE POINT
GRCA	GRAND CANYON
GTB	GEOTECH BORE
MAG	MAG NAIL
MON	MONUMENT
NA	NOT APPLICABLE
NPS	NATIONAL PARK SERVICE

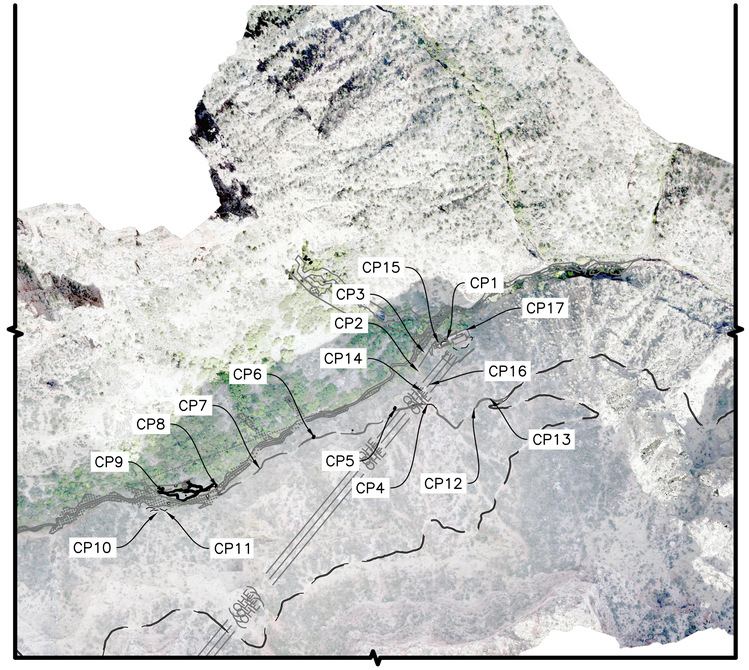
REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 GREGORY YOUNG — STATE OF ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	G. YOUNG	002 C104	GENERAL SURVEY CONTROL POINT TABLE (NORTH RIM)	113 182752
	CADD P. OXBORROW			PMIS/PKG NO. 318719
	TECH. REVIEW: M. GOUGH/R. RHODES		NORTH RIM - GRAND CANYON NATIONAL PARK	SHEET 21 OF 99
	DATE: MAY 2026			

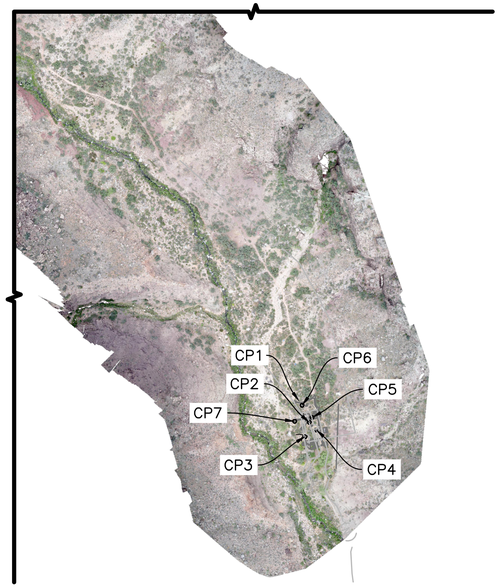
4/23/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\GEN\002-C101.dwg



SUPAI TUNNEL AREA SURVEY CONTROL POINT TABLE				
POINT NO.	NORTHING	EASTING	ELEVATION	DESCRIPTION
XP1	13149291.25	1330843.53	6835.72	SX ON BOULDER
XP2	13149428.22	1330718.32	6873.61	SX
XP3	13149393.57	1330803.47	6844.19	SX
CP400	13149268.04	1330821.00	6830.63	CRP AZS
CP401R	13149347.85	1330848.95	6840.09	CP AZS
TP604	13149310.52	1330792.84	6837.11	NA 60D
LN10085	13149120.70	1330693.79	6931.31	NA 60D



ROARING SPRINGS AREA SURVEY CONTROL POINT TABLE				
POINT NO.	NORTHING	EASTING	ELEVATION	DESCRIPTION
CP1	13142743.19	1335393.45	4859.12	MAG NAIL IN RAILROAD TIE
CP2	13142794.07	1335229.59	4871.45	6" MAG SPIKE
CP3	13142803.82	1335307.32	4863.73	6" MAG SPIKE
CP4	13142704.60	1335126.27	4988.02	X SCRIBED IN ROCK
CP5	13142812.20	1335058.00	4966.12	6" MAG SPIKE
CP6	13143055.77	1334819.05	4967.56	6" MAG SPIKE
CP7	13143198.87	1334615.54	4971.63	6" MAG SPIKE
CP8	13143319.29	1334478.27	4955.44	6" MAG SPIKE
CP9	13143496.19	1334367.86	4982.45	6" MAG SPIKE
CP10	13143493.63	1334283.81	4993.72	6" MAG SPIKE
CP11	13143460.87	1334303.27	4986.94	6" MAG SPIKE
CP12	13142530.20	1335188.98	5032.40	6" MAG SPIKE
CP13	13142489.93	1335251.56	5047.42	6" MAG SPIKE
CP14	13142749.27	1335165.72	4941.80	6" MAG SPIKE
CP15	13142768.72	1335365.64	4866.59	X MARKED IN CONCRETE
CP16	13142730.26	1335200.79	4924.97	6" MAG SPIKE
CP17	13142690.04	1335458.06	4858.87	MAG NAIL



COTTONWOOD CAMPGROUND AREA SURVEY CONTROL POINT TABLE				
POINT NO.	NORTHING	EASTING	ELEVATION	DESCRIPTION
CP1	13134326.78	1333335.22	4039.48	REBAR WITH 2" ALUMINUM CAP
CP2	13134252.27	1333275.14	4031.00	REBAR WITH 2" ALUMINUM CAP
CP3	13134243.57	1333216.79	4028.07	REBAR WITH 2" ALUMINUM CAP
CP4	13134214.98	1333259.28	4028.76	REBAR WITH 2" ALUMINUM CAP
CP5	13134250.50	1333297.48	4032.64	NAIL, 60D
CP6	13134306.78	1333324.09	4037.52	NAIL, 60D
CP7	13134306.67	1333254.13	4031.07	NAIL, 60D

NOTE:
CP1–CP4 CONTROL POINTS ARE 5/8" REBAR WITH ALUMINUM CAPS MARKED "ARIZONA SURVEYING CONTROL", CP5–CP7 ARE 60D NAILS.

300 0 300 600
SCALE OF FEET
1" = 300'

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO.
75262
GREGORY YOUNG
Date Signed 5/11/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: M. GOUCH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 002 C105	TITLE OF SHEET GENERAL OVERALL SURVEY CONTROL CONTROL PLAN (INNER CANYON) NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 22 OF 99	

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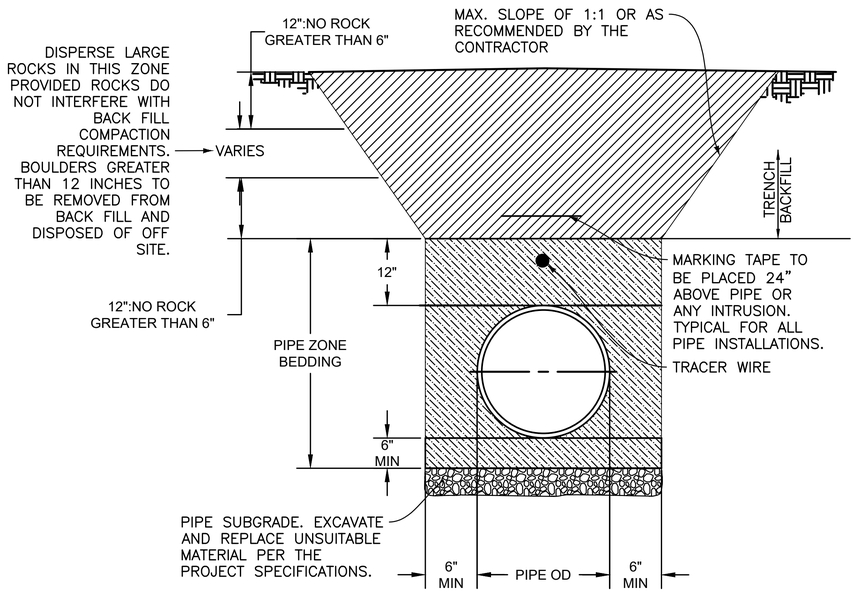
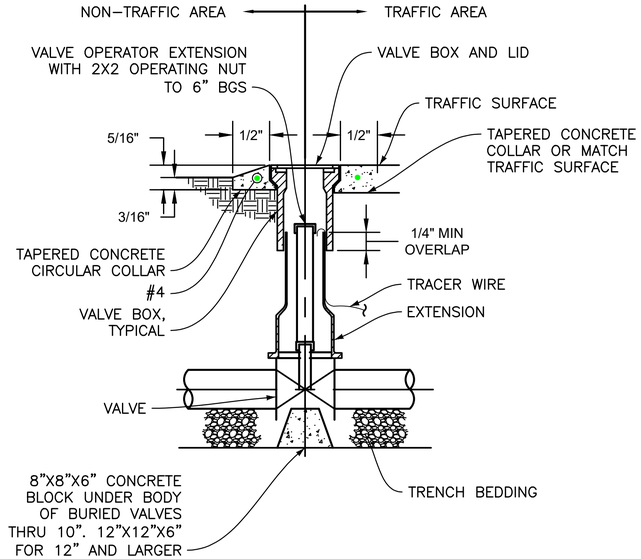
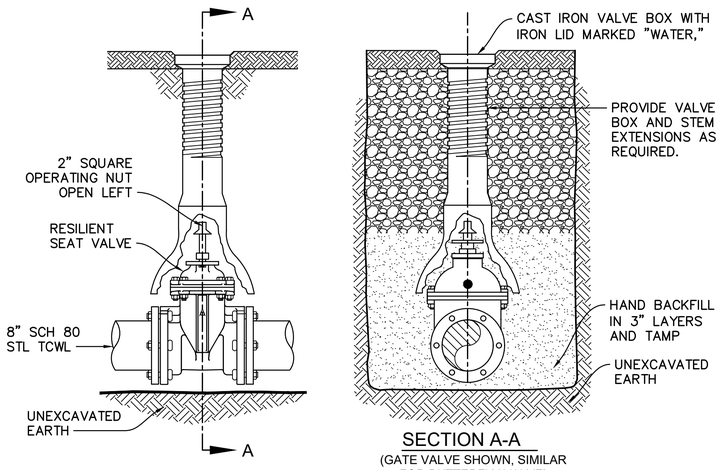


1 002C501 **TYPICAL TRENCH DETAIL**
NOT TO SCALE

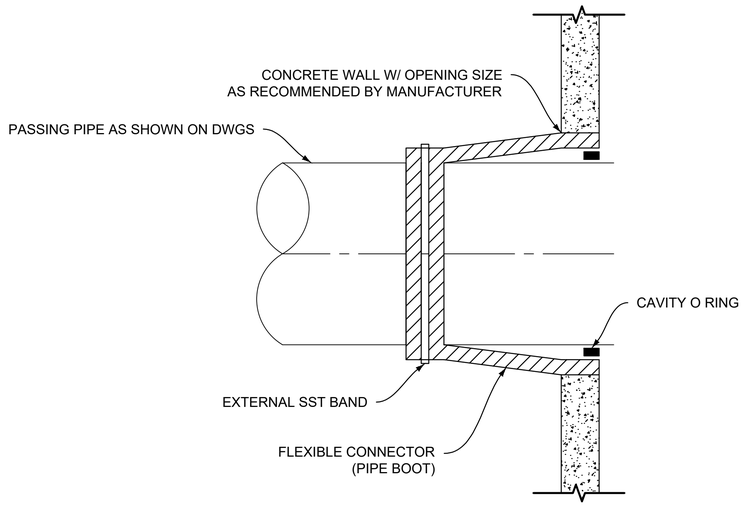


3 002C501 **BURIED VALVE BOX W/ TRACER WIRE**
NOT TO SCALE



SECTION A-A
(GATE VALVE SHOWN, SIMILAR FOR BUTTERFLY VALVE)

2 002C501 **GATE VALVE DETAIL**
NOT TO SCALE

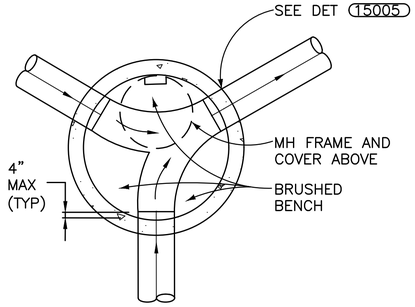


4 002C501 **WALL PENETRATION**
NOT TO SCALE

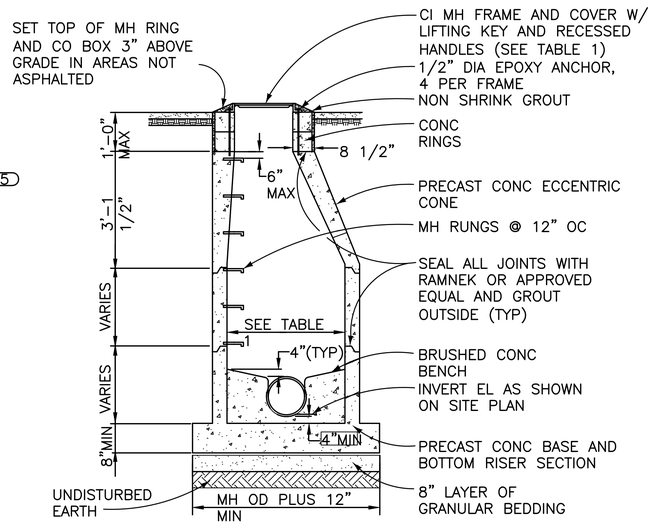
REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75282 — GREGORY YOUNG — Date Signed — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD C. WITTENMEIER TECH. REVIEW: M. GOUCH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 002 C501	TITLE OF SHEET GENERAL CIVIL DETAILS – UTILITIES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 23 OF 99

4/17/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\GEN\002-C501.dwg

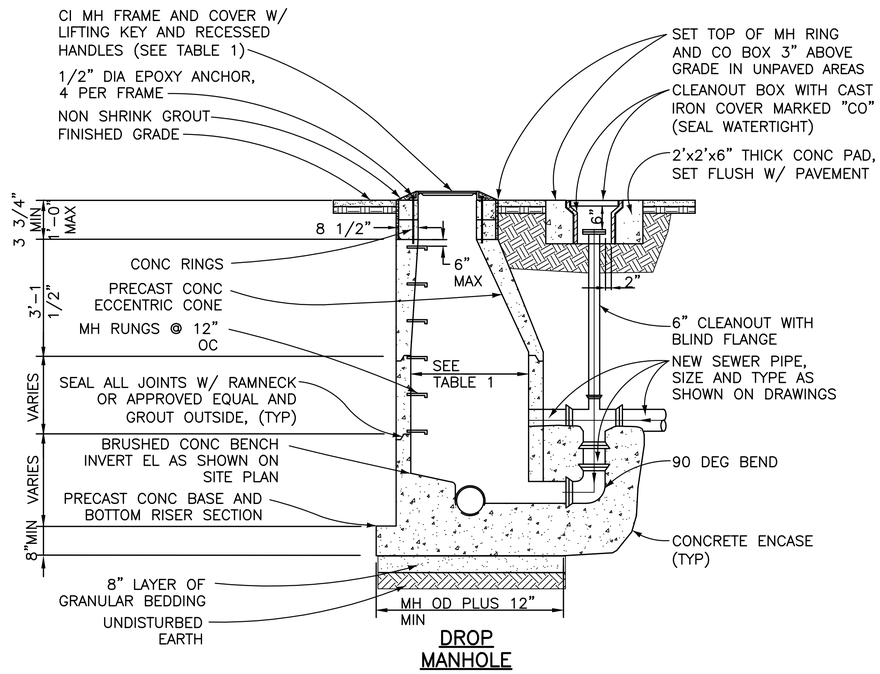


TYPICAL PLAN



STANDARD MANHOLE

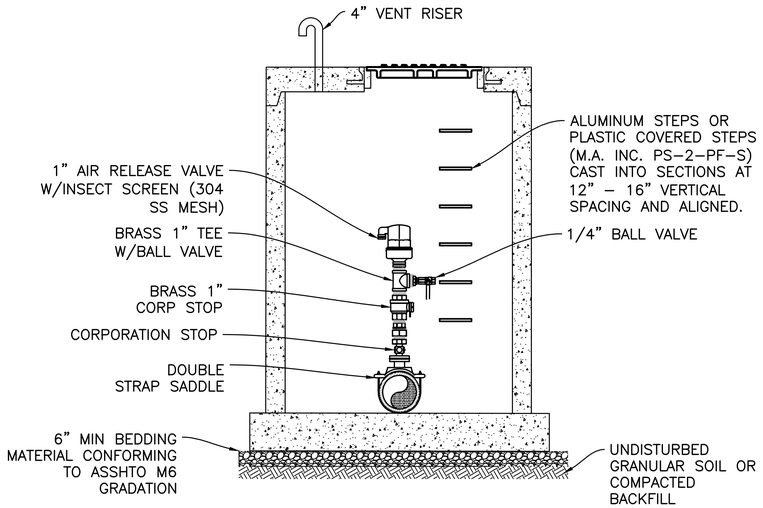
TABLE 1		
PIPE DIA	MH BARREL DIA	MH FRAME AND COVER DIA
12" OR LESS	4' DIA	24" DIA
14" TO 30"	5' DIA	30" DIA



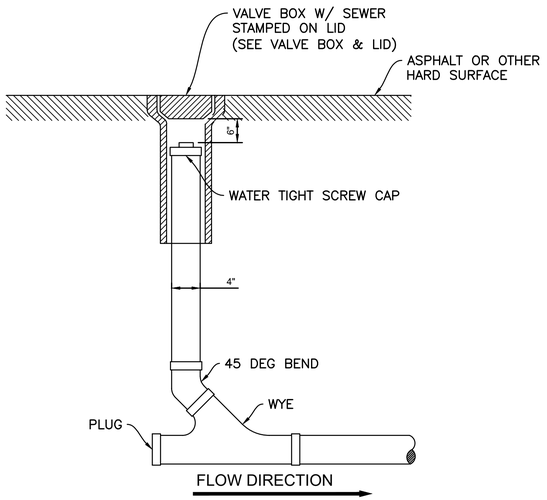
DROP MANHOLE

NOTES:
1. ALL COMPONENTS SHALL BE CAPABLE OF HS-20 LOADING.
2. INSTALL ALL PIPES INTO MANHOLES WITH FLEXIBLE PIPE-TO-MANHOLE CONNECTORS.
3. DROP MH SHALL BE USED FOR ALL SITUATIONS WHERE PIPES ENTER MH AT 24" OR HIGHER ABOVE BENCH.
4. MIN OF 18" CONC REQUIRED BENEATH 90° BEND.

1 / 002C502 MANHOLE W/ FORMED BENCH
NOT TO SCALE



2 / 002C502 DISTRIBUTION AIR RELEASE VALVE ASSEMBLY
NOT TO SCALE

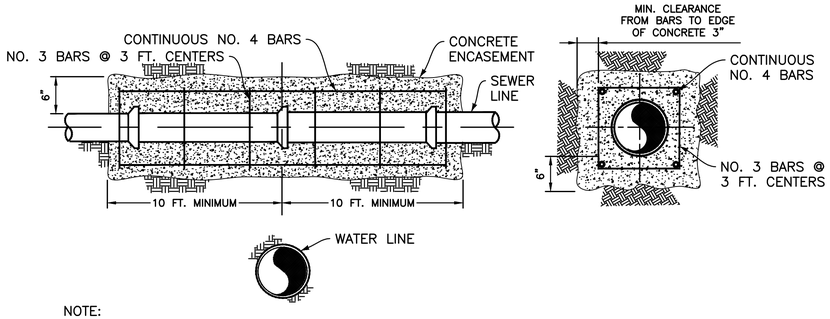


3 / 002C502 SANITARY SEWER CLEANOUT W/ VALVE BOX & LID
NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 GREGORY YOUNG — Date Signed 5/1/2026 — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: R. FREDERICKS DRAWN: C. WITTENMEIER TECH. REVIEW: M. GOUCH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 002 C502	TITLE OF SHEET GENERAL CIVIL DETAILS – UTILITIES CONTINUED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 24 OF 99

4/17/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\GEN\002-C501.dwg

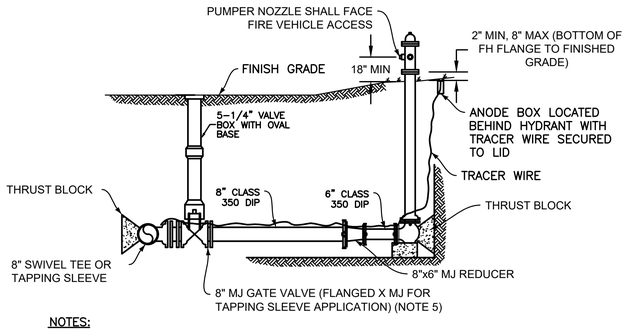


NOTE:

CONCRETE ENCASEMENT IS REQUIRED WHERE CALLED OUT IN THE CONSTRUCTION DOCUMENTS AND IS REQUIRED WHEN CLEAR VERTICAL DISTANCE BETWEEN SANITARY SEWER AND WATER LINES IS LESS THAN 1'–6" OR HORIZONTAL DISTANCE IS LESS THAN 6', PER ADEQ REQUIREMENTS. BETWEEN PARALLEL LINES WITH LESS THAN 6 FT OF SEPARATION. CONCRETE ENCASEMENT OF SANITARY SEWER IS REQUIRED IN ALL CASES WHERE SEWER LINE IS ABOVE WATER LINE.

1 / 002C503 CONCRETE ENCASEMENT DETAIL

NOT TO SCALE

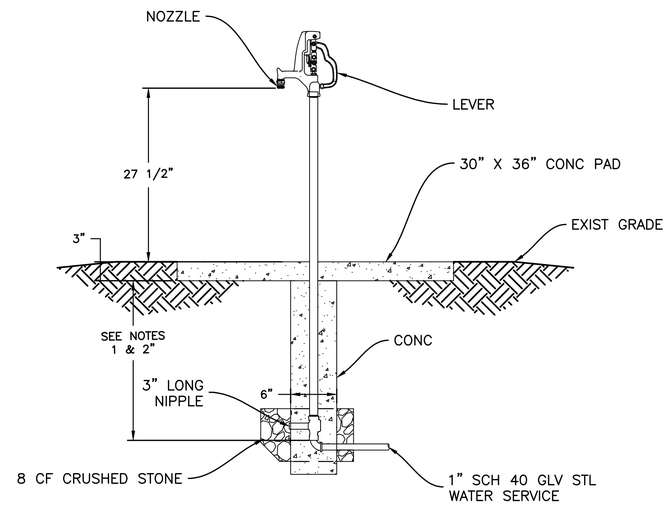


NOTES:

1. ALL FITTINGS, DI PIPE, AND HYDRANT BARREL SHALL BE POLYETHYLENE WRAPPED PER AWWA C–105. PROVIDE PERFORATION AT HYDRANT DRAIN HOLES.
2. TREES, SHRUBS, AND STRUCTURES SHALL NOT BE LOCATED WITHIN THE UTILITY EASEMENT OR RIGHT–OF–WAY FOR THE HYDRANT. HYDRANT UTILITY EASEMENTS AND RIGHTS–OF–WAY SHALL BE 10–FOOT WIDE (MIN), AND EXTEND 5–FEET BEHIND THE HYDRANT. VARIANCES FROM THIS ENCROACHMENT CRITERIA WILL REQUIRE THE WRITTEN APPROVAL OF THE FIRE MARSHALL.
3. PRIOR TO PUTTING THE HYDRANT INTO SERVICE THE FOLLOWING SHALL BE PERFORMED: NOZZLE THREADS SHALL BE GREASED, OIL RESERVOIR SHALL BE FILLED, AND HYDRANTS SHALL BE PAINTED "RUSTOLEUM" HIGH PERFORMANCE ENAMEL (#7564, SAFETY RED).
4. THE ENTIRE PIPING ASSEMBLY (HYDRANT SHOE TO CONNECTION AT MAIN) SHALL BE RESTRAINED.
5. THE VALVE NUT SHALL HAVE A MAXIMUM BURIED DEPTH OF 4' FROM FINISHED GRADE.
6. WHERE FIRE HYDRANTS ARE SUBJECT TO IMPACT BY A MOTOR VEHICLE, BOLLARD POSTS OR OTHER APPROVED MEANS SHALL BE PROVIDED.

2 / 002C503 FIRE HYDRANT INSTALLATION

NOT TO SCALE



NOTES:

1. NORTH RIM MIN. BURY DEPTH 42"
2. INNER CANYON MIN. BURY DEPTH 24"

3 / 002C503 FROST RESISTANT YARD HYDRANT

NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — Date Signed 5/11/26 — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: C. WITTENMEIER TECH. REVIEW: M. GOUCH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 002 C503	TITLE OF SHEET GENERAL CIVIL DETAILS – UTILITIES CONTINUED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 25 OF 99

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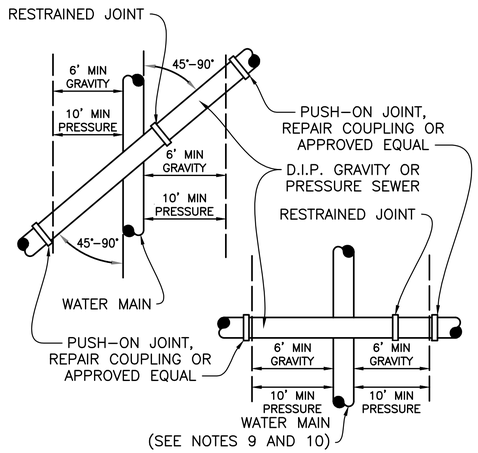
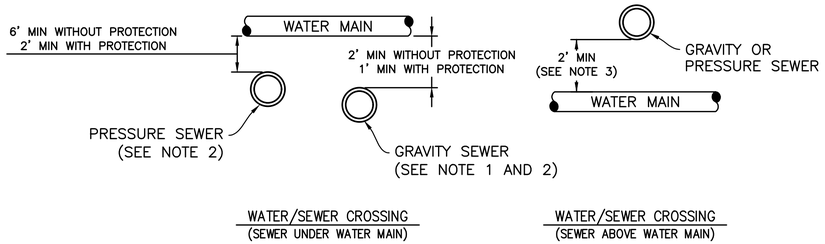


SEWER/WATER SEPARATION
(PARALELL ALIGNMENTS)

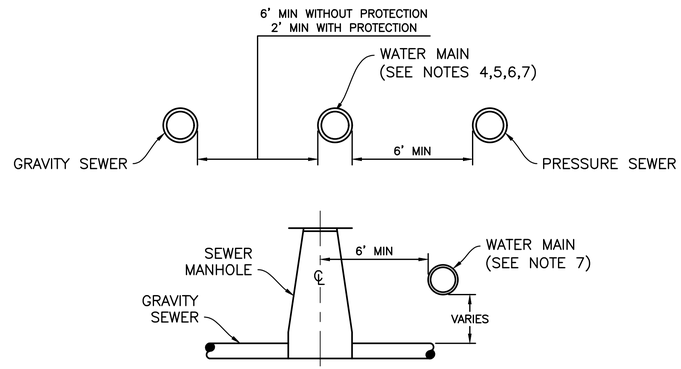
PLAN VIEWS



WATER/SEWER CROSSING
(SEWER UNDER WATER MAIN)

WATER/SEWER CROSSING
(SEWER ABOVE WATER MAIN)

VERTICAL SEPARATION



HORIZONTAL SEPARATION

1 / 002C504 WATER/SEWER CROSSING DETAILS
SCALE: NOT TO SCALE

GENERAL NOTES:

1. WHERE A WATERMAIN CROSSES ABOVE A GRAVITY SEWER WITH A 2' OR GREATER VERTICAL CLEARANCE (OUTSIDE SURFACE TO OUTSIDE SURFACE), NO EXTRA PROTECTION IS REQUIRED. WHERE A WATERMAIN CROSSES ABOVE A PRESSURE SEWER WITH 2' OR GREATER VERTICAL CLEARANCE, NO EXTRA PROTECTION IS REQUIRED.
2. WHERE A WATERMAIN MUST CROSS ABOVE A GRAVITY SEWER WITH LESS THAN 2' OF CLEARANCE OR A PRESSURE SEWER WITH LESS THAN 6' OF CLEARANCE, CONSTRUCT OR REPLACE THE SEWER LINE WITH EPOXY COATED DUCTILE IRON PIPE (D.I.P.) OR APPROVED EQUAL. THE D.I.P. MUST HAVE A "PUSH-ON" JOINT OR APPROVED EQUAL. IF THE SEWER JOINTS ARE LOCATED 6' OR LESS BEYOND THE OUTSIDE SURFACE OF THE WATERMAIN, THEN A RESTRAINED JOINT OR APPROVED EQUAL SHALL BE USED. SEE THE PLAN VIEWS FOR REFERENCE. IN NO CASE SHALL THE WATERMAIN BE LESS THAN 1' ABOVE A GRAVITY SEWER OR 2' ABOVE A PRESSURE SEWER.
3. WHERE A WATERMAIN CROSSES BELOW EITHER A GRAVITY SEWER OR A PRESSURE SEWER, CONSTRUCT OR REPLACE THE SEWER LINE WITH D.I.P. OR APPROVED EQUAL FOLLOWING THE GUIDELINES GIVEN IN NOTE NO. 2. IN NO CASE SHALL THE WATERMAIN BE LESS THAN 2' BELOW EITHER A GRAVITY OR PRESSURE SEWER LINE.
4. THE MINIMUM HORIZONTAL CLEARANCE WITHOUT EXTRA PROTECTION BETWEEN A WATERMAIN AND A PRESSURE OR GRAVITY SEWER LINE SHALL BE 6' OUTSIDE SURFACE TO OUTSIDE SURFACE.
5. WHERE A 6' HORIZONTAL CLEARANCE CANNOT BE MAINTAINED WITH A GRAVITY SEWER, CONSTRUCT OR REPLACE THE SEWER LINE WITH D.I.P. OR APPROVED EQUAL FOLLOWING THE GUIDELINES GIVEN IN NOTE NO. 2. IN NO CASE SHALL A GRAVITY SEWER LINE BE LOCATED LESS THAN 2' HORIZONTALLY FROM A WATERMAIN.
6. THE MINIMUM HORIZONTAL CLEARANCE BETWEEN A WATERMAIN AND A SEWER MANHOLE SHALL BE 6' FROM THE OUTSIDE SURFACE OF THE WATERMAIN TO THE CENTERLINE OF THE MANHOLE.
7. IN NO CASE SHALL A PRESSURE SEWER LINE BE LOCATED LESS THAN 6' HORIZONTALLY FROM A WATER MAIN.
8. THE AMERICAN NATIONAL STANDARD FOR THE THICKNESS DESIGN OF DUCTILE IRON PIPE (ANSI/AWWA C150/A21.50) SHALL BE USED TO DETERMINE THE REQUIRED CLASS OF D.I.P. EXCEPT THAT FOR D.I.P. WITH A DIAMETER OF 3" THROUGH 24", A MINIMUM OF CLASS 350 IS REQUIRED AND FOR D.I.P. WITH A DIAMETER GREATER THAN 24", A MINIMUM OF CLASS 200 IS REQUIRED.
9. WHERE THE SEWER LINE IS A GRAVITY SEWER, THE D.I.P. INSTALLATION OF REPLACEMENT PIPE SHALL EXTEND A MINIMUM OF 6' BEYOND EACH SIDE OF THE WATERMAIN, MEASURED HORIZONTALLY FROM AND PERPENDICULAR TO THE WATERMAIN.
10. D.I.P. INSTALLATION OF REPLACEMENT PIPE SHALL EXTEND A MINIMUM OF 10' BEYOND EACH SIDE OF THE WATERMAIN, MEASURED HORIZONTALLY FROM AND PERPENDICULAR TO THE WATERMAIN.
11. ARIZONA DEPARTMENT OF ENVIRONMENTAL QUALITY APPROVAL FOR ALTERNATE CONSTRUCTION TECHNIQUES AND JOINTS IS NECESSARY WHEN UNUSUAL CONDITIONS ARE ENCOUNTERED. THIS AUTHORIZATION MUST BE ON A CASE BY CASE BASIS, AND ACCEPTANCE OF ONE INSTANCE DOES NOT GUARANTEE ACCEPTANCE OF THE SAME SOLUTION A DIFFERENT INSTANCE.

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: R. FREDERICKS DRAWN: C. WITTENMEIER TECH. REVIEW: M. COUGH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 002 C504	TITLE OF SHEET GENERAL CIVIL DETAILS – UTILITIES CONTINUED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 26 OF 99

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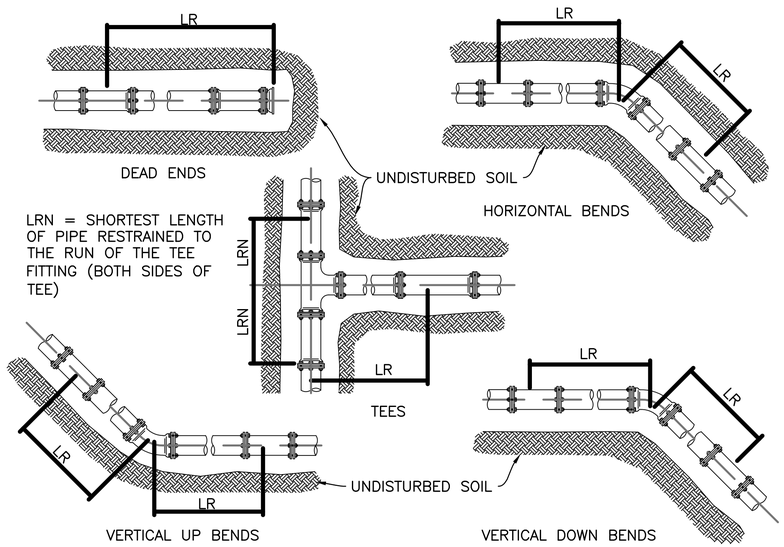


1 / 002C505 JOINT RESTRAINT DETAIL–1

NOT TO SCALE

RESTRAINED LENGTHS, LR, FOR DUCTILE IRON PIPE								
PIPE SIZE	HORIZONTAL BENDS				VERTICAL BENDS			DEAD ENDS
	11 1/4°	22 1/2°	45°	90°	11 1/4°	22 1/2°	45°	
4"	3'	6'	13'	31'	11'	22'	46'	112'
6"	4'	8'	18'	42'	15'	31'	65'	157'
8"	5'	11'	23'	54'	20'	41'	84'	204'
12"	7'	15'	31'	74'	28'	57'	118'	284'

RESTRAINED LENGTHS, LR, FOR DUCTILE IRON PIPE		
TEE SIZE	LRN=0'	LRN=10'
	LENGTH OF RESTRAINED BRANCH	
4"x4"	74'	21'
8"x4"	74'-0"	1'-0"
8"x8"	135'-0"	80'-0"

RESTRAINED LENGTHS, LR, FOR PVC PIPE								
PIPE SIZE	HORIZONTAL BENDS				VERTICAL BENDS			DEAD ENDS
	11 1/4°	22 1/2°	45°	90°	11 1/4°	22 1/2°	45°	
4"	3'	5'	11'	27'	7'	15'	31'	74'
6"	4'	7'	15'	37'	10'	21'	43'	104'
8"	5'	10'	20'	48'	13'	27'	56'	135'
12"	6'	13'	27'	65'	19'	38'	78'	189'

RESTRAINED LENGTHS, LR, FOR PVC PIPE		
TEE SIZE	LRN=0'	LRN=10'
	LENGTH OF RESTRAINED BRANCH	
4"x4"	112'	32'
8"x4"	112'-0"	1'-0"
8"x8"	204'-0"	121'-0"

NOTES:

1. ALL JOINTS WITHIN THE SPECIFIED LENGTH LR MUST BE RESTRAINED. ALL LENGTHS ARE GIVEN IN FEET.
2. THE MAXIMUM TEST PRESSURE SHALL NOT EXCEED 200 PSI.
3. THE MINIMUM DEPTH OF BURY SHALL BE 2' TO TOP OF PIPE.
4. RESTRAINED LENGTHS MAY BE REDUCED WHEN SUPPORTED BY ENGINEERING CALCULATIONS.
5. ALL DUCTILE IRON IS POLYETHYLENE ENCASED.
6. THE SOIL TYPE IS "GP" AS DEFINED BY THE UNITED SOIL CLASSIFICATIONS, ASTM STANDARD D2487.
7. THE TRENCH IS TYPE 5, AS PER ANSI/AWWA C150/A21.5, TRENCH CONDITIONS.

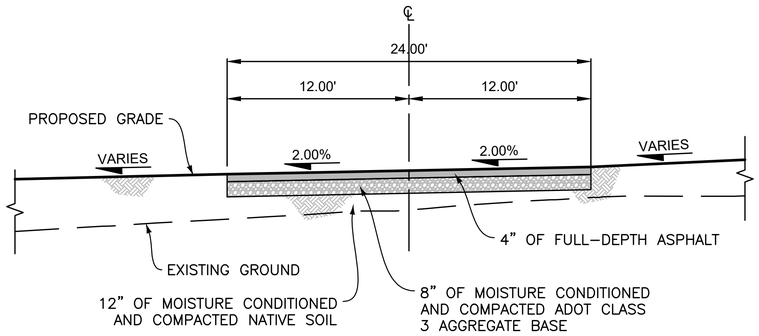
2 / 002C505 JOINT RESTRAINT DETAIL–2

NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — ARIZONA, U.S.A. — 4/17/26

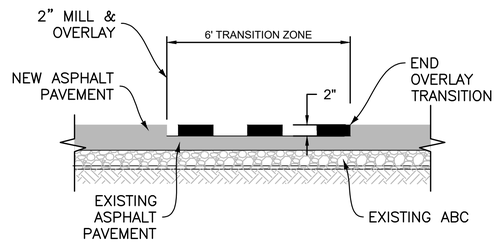
A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	G. YOUNG CADD C. WITTENMEIER TECH. REVIEW: M. COUGH/R. RHODES DATE: MAY 2026	002 C505	GENERAL CIVIL DETAILS – JOINTS NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 27 OF 99

4/17/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\GEN\002-C501.dwg



1 002C506 **TYPICAL ASPHALT ROAD**

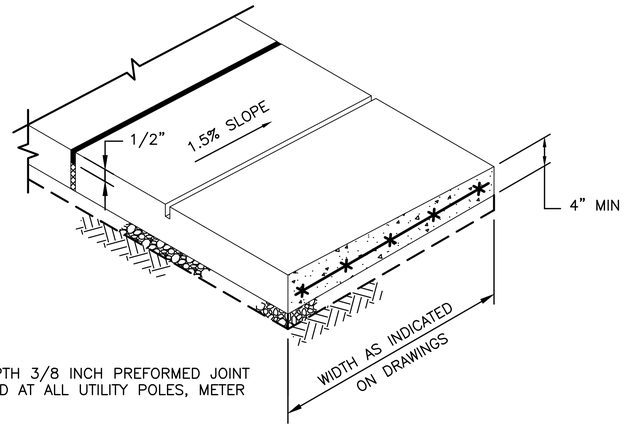
NOT TO SCALE



NOTE:
TRANSITION BETWEEN NEW HOT MIX ASPHALT AND EXISTING ASPHALT PAVEMENT.

2 002C506 **ASPHALT PAVEMENT TRANSITION**

NOT TO SCALE

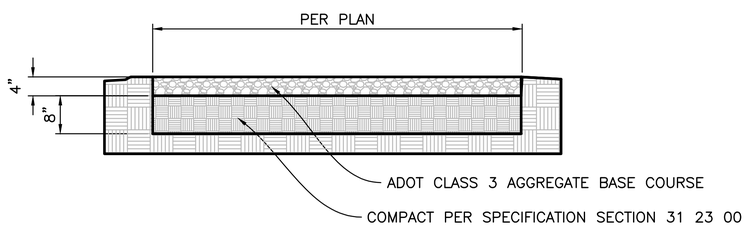


NOTES:

1. PROVIDE FULL DEPTH 3/8 INCH PREFORMED JOINT FILLER ALL AROUND AT ALL UTILITY POLES, METER BOXES, ETC.
2. PLACE PREFORMED JOINT FILLER FULL DEPTH THRU JOINTS AT POINT OF TANGENCY AND AT MINIMUM 25 FOOT INTERVALS.
3. THICKNESS OF CONCRETE AS SHOWN ON DRAWINGS.

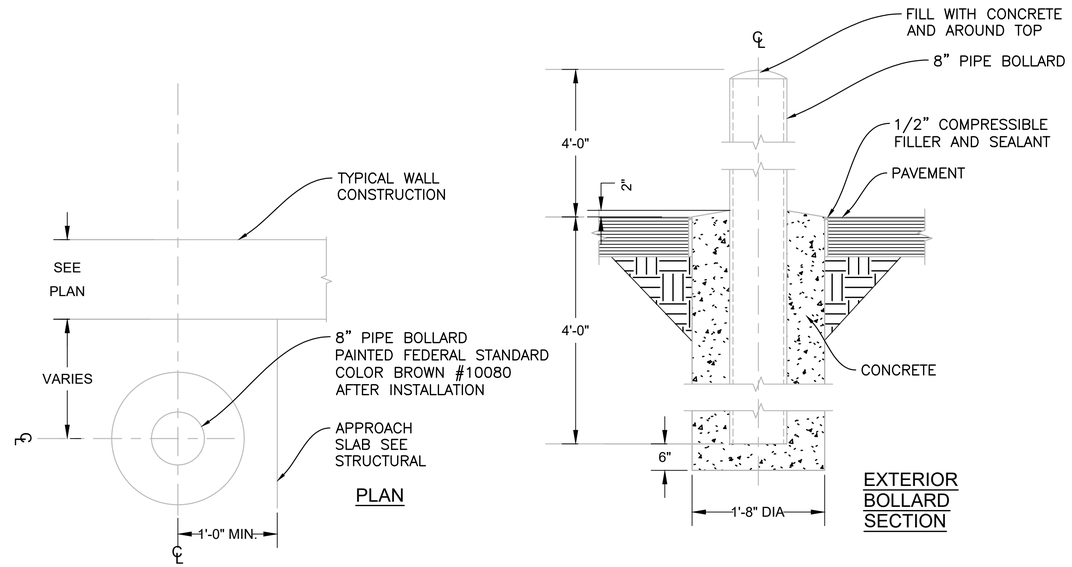
3 002C506 **TYPICAL SIDEWALK**

NOT TO SCALE



4 002C506 **TYPICAL GRAVEL TRAIL RECONSTRUCTION**

NOT TO SCALE



5 002C506 **EXTERIOR BOLLARD DETAILS**

NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — Date Signed 5/11/26 — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD C. WITTENMEIER TECH. REVIEW: M. GOUCH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 002 C506	TITLE OF SHEET GENERAL CIVIL DETAILS – PAVING NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 28 OF 99

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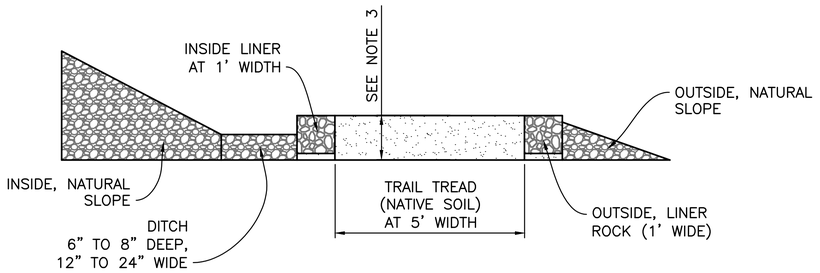
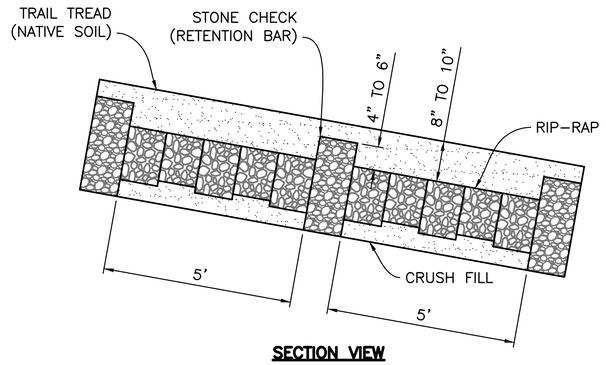


SECTION VIEW

NOTE

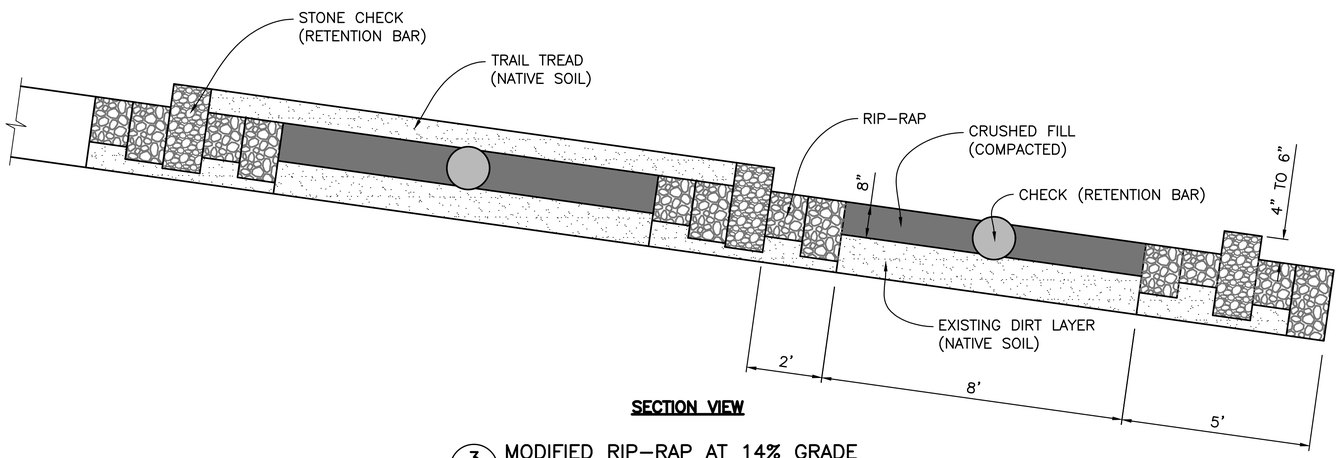
1. THE FOLLOWING DESIGN ILLUSTRATES A COMPLETED SECTION OF TRAIL REGARDLESS OF THE TRAIL GRADE.
2. REFER TO TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL DETAILS AND MATERIAL REQUIREMENTS.
3. SEE TRAIL SECTION VIEWS FOR TRAIL DEPTH REQUIREMENTS.
4. ALL TRAILS IMPACTED BY CONSTRUCTION SHALL BE RESTORED TO PRE–CONSTRUCTION CONDITION OR BETTER.

1 / 002C507 COMPLETED TRAIL GRADE
NOT TO SCALE



SECTION VIEW

2 / 002C507 RIP–RAP FOR GRADES 18% OR HIGHER
NOT TO SCALE



SECTION VIEW

3 / 002C507 MODIFIED RIP–RAP AT 14% GRADE
NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — Date Signed — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD C. WITTENMEIER TECH. REVIEW: J. GLOVER DATE: MAY 2026	SUB SHEET NO. 002 C507	TITLE OF SHEET GENERAL CIVIL DETAILS – GRADING NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 29 OF 99

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GENERAL EROSION CONTROL NOTES

A. THE FOLLOWING REQUIREMENTS SHALL BE THE OBLIGATION OF THE CONTRACTOR, UNTIL SUCH TIME AS THE PLAN IS CERTIFIED AS PROPERLY COMPLETED, OR UNTIL SUCH TIME AS OTHERWISE ALLOWED BY THE NPS TO BE VOIDED, MODIFIED, OR REPLACED.

B. ADDITIONAL EROSION CONTROL MEASURES MAY BE REQUIRED DUE TO CHANGED IN CONSTRUCTION, UNFORESEEN EROSION PROBLEMS, OR IF THE CONSTRUCTION PLANS DO NOT FUNCTION AS INTENDED.

C. INSTALL ALL SOIL EROSION CONTROL MEASURES IN ACCORDANCE WITH THE AUTHORIZATION TO DISCHARGE UNDER THE NATIONAL POLLUTANT DISCHARGE ELIMINATION SYSTEM (NPDES) AS REGULATED BY THE ARIZONA DEPARTMENT OF ENVIRONMENTAL QUALITY (ADEQ).

D. A STATE ISSUED STORMWATER CONSTRUCTION PERMIT AND SWMP (STORM WATER MANAGEMENT PLAN) IS REQUIRED AND MUST BE MAINTAINED THROUGH THE DURATION OF THE PROJECT. THE SWMP IS HELD BY THE CONTRACTOR AND A COPY WILL BE SUPPLIED TO NPS.

E. DURING CONSTRUCTION, THE CONTRACTOR SHALL BE RESPONSIBLE FOR INSTALLING AND MAINTAINING ALL EROSION CONTROL FACILITIES SHOWN IN DRAWINGS AND REMOVAL WHEN DIRECTED BY THE NPS.

F. ALL PERIMETER EROSION CONTROL MEASURES SHALL BE INSTALLED AND FUNCTIONAL, PRIOR TO ANY OTHER EARTH-DISTURBING ACTIVITY. ALL OTHER STRUCTURAL EROSION CONTROL MEASURES SHALL BE IMPLEMENTED AS SOON AS THE FACILITIES, AROUND WHICH THEY ARE BASED, BECOME OPERATIONAL.

G. ANY EROSION CONTROL FACILITY DAMAGED OR DESTROYED PREMATURELY, BY ANY MEANS, SHALL BE IMMEDIATELY REPAIRED BY THE CONTRACTOR.

H. WHERE TOPSOIL IS REPLACED, IT SHALL BE INSTALLED WITH THE TOP OF THE TOPSOIL SURFACE SET AT FINISH GRADE.

I. TEMPORARY EROSION CONTROL MEASURES SHALL NOT BE REMOVED UNTIL SUCH TIME AS ALL TRIBUTARY DISTURBED AREAS ARE SUFFICIENTLY ESTABLISHED IN THE OPINION OF THE NPS TO MINIMIZE EROSION POTENTIAL.

J. CONTRACTOR SHALL APPLY FOR AND PROCURE A FEDERAL JURISDICTION STORMWATER COMPLIANCE PERMITS.

K. CONTRACTOR SHALL DETERMINE LOCATIONS OF STABILIZED STAGING AREA(S) AND CONCRETE WASHOUT(S) CONTRACTOR SHALL GET APPROVAL FROM NPS PRIOR TO IMPLEMENTATION.

L. CONTRACTOR SHALL REMOVE ALL SEDIMENT, MUD, AND CONSTRUCTION DEBRIS THAT ACCUMULATES IN THE FLOW LINES, RESIDENTIAL AREAS, AND PUBLIC RIGHT OF WAYS AS A RESULT OF THIS CONSTRUCTION PROJECT. REMOVAL SHALL BE CONDUCTED WITHIN 24-HOURS OR AS DIRECTED BY THE EROSION CONTROL INSPECTOR. FAILURE TO DO SO WITHIN THE 24 HOUR PERIOD WILL RESULT IN BEING SUSPENDED UNTIL ALL DEFICIENCIES ARE CORRECTED. NO CONTRACT EXTENSION OR COMPENSATION WILL BE ALLOWED FOR THIS PERIOD OF SUSPENSION.

M. CONTRACTOR SHALL PREVENT LOSS OF CUT AND FILL MATERIAL BEING TRANSPORTED TO AND FROM THE SITE BY TAKING APPROPRIATE MEASURES. ALL MUD AND SEDIMENT TRACKED ONTO PUBLIC STREETS SHALL BE CLEANED IMMEDIATELY BY CONTRACTOR. STREET CLEANING INCLUDES SHOVELING AND SWEEPING ACTIVITIES. AT NO TIME SHALL SEDIMENT BE WASHED DOWN INLETS INTO NEARBY STORM SEWER SYSTEMS.

N. SOILS THAT WILL BE STOCKPILED FOR MORE THAN 30 DAYS SHALL BE MULCHED AND SEEDED WITH A TEMPORARY OR PERMANENT GRASS COVER WITHIN 7 DAYS OF STOCKPILE CONSTRUCTION.

O. TOPSOIL SHALL BE SAVED AND NOT REMOVED FROM THE SITE EXCEPT AS SET FORTH IN THESE DRAWINGS. TOPSOIL AND OVERBURDEN SHALL BE SEGREGATED AND STOCKPILED SEPARATELY. SUITABLE OVERBURDEN AND THE TOPSOIL SHALL BE REDISTRIBUTED WITHIN THE GRADED AREA AFTER ROUGH GRADING TO PROVIDE A SUITABLE BASE FOR AREAS WHICH WILL BE VEGETATED

RUNOFF FROM STOCKPILE SHALL BE CONTROLLED TO PREVENT SEDIMENT ENTERING THE RECEIVING WATERS,

P. CONTRACTOR SHALL IMPLEMENT EROSION SEDIMENT CONTROL BEST MANAGEMENT PRACTICES (BMPS) AT THE BEGINNING OF CONSTRUCTION ACTIVITIES. BMPS SHALL BE MAINTAINED AND FULLY FUNCTIONAL FOR THE CONTRACT WARRANTY PERIOD.

Q. INSPECTION AND MAINTENANCE OF EROSION AND SEDIMENT CONTROL BMPS ARE THE CONTINUOUS OBLIGATIONS OF THE CONTRACTOR. CONTRACTOR SHALL MAINTAIN EROSION AND SEDIMENT CONTROL MEASURES (ESCMS) WEEKLY AND AFTER EVERY PRECIPITATION EVENT. CONTRACTOR SHALL MAINTAIN ESCMS TO THE SATISFACTION OF THE NPS. FAILURE TO CORRECT DEFICIENCIES WITHIN THE 24-HOUR PERIOD WILL RESULT IN THE PROJECT BEING SUSPENDED UNTIL ALL DEFICIENCIES ARE CORRECTED. NO CONTRACT EXTENSION OR COMPENSATION WILL BE ALLOWED FOR THIS PERIOD OF SUSPENSION. THE INSPECTION REPORTS SHALL BE MADE AVAILABLE TO THE NPS UPON REQUEST.

R. ACCUMULATED SEDIMENT AND DEBRIS SHALL BE REMOVED FROM AN ESCMS WHEN THE SEDIMENT LEVEL REACHES ONE HALF THE HEIGHT OF THE ESCMS, AT ANYTIME THAT SEDIMENT OR DEBRIS ADVERSELY IMPACTS THE FUNCTIONING ESCMS.

S. THE CLEANING OF CONCRETE TRUCK DELIVERY CHUTES, EXCEPT IN DESIGNATED AREAS, IS PROHIBITED AT THE JOB SITE.

T. ALL MATERIAL IMPORTED TO OR EXPORTED FROM THE SITE SHALL BE PROPERLY COVERED TO PREVENT THE LOSS OF MATERIAL DURING TRANSPORT.

U. FUGITIVE DUST EMISSIONS RESULTING FROM GRADING ACTIVITIES AND/OR WIND SHALL BE CONTROLLED USING REASONABLY AVAILABLE CONTROL TECHNOLOGY AS DEFINED BY THE ARIZONA DEPARTMENT OF PUBLIC HEALTH AND ENVIRONMENT.

V. THE CONTRACTOR SHALL NOTIFY THE NPS EROSION CONTROL INSPECTOR UPON INSTALLATION OF SEDIMENT AND EROSION CONTROL MEASURES 24 HOURS PRIOR TO STARTING GRADING WORK.

W. EROSION AND SEDIMENT CONTROL MEASURES SHALL BE IN PLACE AND INSPECTED BY THE NPS PRIOR GRADING ACTIVITIES. ADDITIONAL EROSION AND SEDIMENT CONTROL MEASURES MAY BE REQUIRED DURING AND AFTER CONSTRUCTION AND SHALL BE INSTALLED WITHIN 48 HOURS OF NOTIFICATION BY THE NPS. THE CONTRACTOR SHALL PLAN, INSTALL, AND MAINTAIN ALL EROSION CONTROL MEASURES AS INDICATED ON THE PLANS AND AS NECESSARY TO PREVENT SEDIMENT DEPOSITION OFF-SITE.

X. CONTRACTOR SHALL PROVIDE SILT FENCE (SF) AROUND WORK.

Y. CONTRACTOR SHALL PROVIDE VEHICLE TRACKING CONTROLS (VTC) AT EACH LOCATION WHERE THE CONTRACTOR ENTERS PAVED STREETS FROM OPEN FIELDS OR AS INDICATED ON THE PLANS.

Z. FOR MORE INFORMATION SEE SPECIFICATION 31 25 00 SOIL EROSION AND SEDIMENT CONTROL.

ARIZONA NPDES GENERAL PERMIT

STORM WATER POLLUTION PLAN CHECKLIST

DOCUMENTATION AND REPORTING REQUIREMENTS:
A COPY OF CGP-2020 (AZG2020-001)
A COPY OF THE NOI SUBMITTED TO ADEQ, INCLUDING ANY CORRESPONDENCE RELATED TO COVERAGE UNDER THIS PERMIT.
A COPY OF THE AUTHORIZATION CERTIFICATE FROM ADEQ.
IDENTIFICATION OF ANY MUNICIPALITY THAT RECEIVED A COPY OF THE AUTHORIZATION CERTIFICATE
COPIES OF ANY OTHER ENVIRONMENTAL AGREEMENTS
DESCRIPTION S / DATES OF SIGNIFICANT SPILLS, LEAKS OR OTHER RELEASES
DOCUMENTATION OF REPAIRS INCLUDING DATE(S) OF DISCOVERY AND REPAIRS MADE
BUFFER ALTERNATIVE SELECTED FOR ACTIVITIES LOCATED WITHIN 50 FEET OF A PERENNIAL WATER
DOCUMENTATION TO SUPPORT THE CHANGE OF ACTIVE TO INACTIVE UNSTAFFED
DESCRIPTION OR PERMANENT, POST-CONSTRUCTION STORMWATER MANAGEMENT CONTROL MEASURES

INSPECTIONS, MAINTENANCE AND CORRECTIVE ACTION:
IDENTIFY QUALIFIED PERSONNEL RESPONSIBLE FOR INSPECTIONS
IDENTIFY INSPECTION SCHEDULE
INSPECTION AND MAINTENANCE CHECKLISTS
DESCRIPTION OR CORRECTIVE ACTION TAKEN
CORRECTIVE ACTION REPORTS

INSPECTION REPORTS SUMMARIZING:
INSPECTION DATE
NAME AND TITLE OF INSPECTOR
FOR INSPECTIONS OCCURRING DURING OR AFTER A STORM EVENT:
A. BEST ESTIMATE OF THE BEGINNING OF EACH STORM EVENT;
B. DURATION OF EACH EVENT;
C. APPROXIMATE AMOUNT OF RAINFALL FOR EACH EVENT (IN INCHES);
D. A DESCRIPTION OF THE PHYSICAL CHARACTERISTICS OF THE STORMWATER DISCHARGE FROM THE SITE, WHEN PRESENT;
E. DOCUMENT THE EVIDENCE OF EROSION, SEDIMENTATION, AND OTHER POLLUTANTS;
F. DOCUMENT THE PRESENCE OF CONTROL MEASURE IN ALL AREAS INSPECTED AND WHETHER SUCH CONTROL ARE OPERATING EFFECTIVELY.
MEASURES / AREAS INSPECTED
OBSERVED CONDITIONS
SOURCES OF NON-STORMWATER DISCHARGES OCCURRING
MATERIAL STORAGE AREAS
CORRECTIVE ACTIONS REQUIRED
NECESSARY CHANGES TO THE SWPP AND IMPLEMENTATION DATES
INSTANCES OF NON-COMPLIANCE

ANY OPERATOR SIGNING DOCUMENTS UNDER THE TERMS OF THIS PERMIT SHALL MAKE THE FOLLOWING CERTIFICATION:

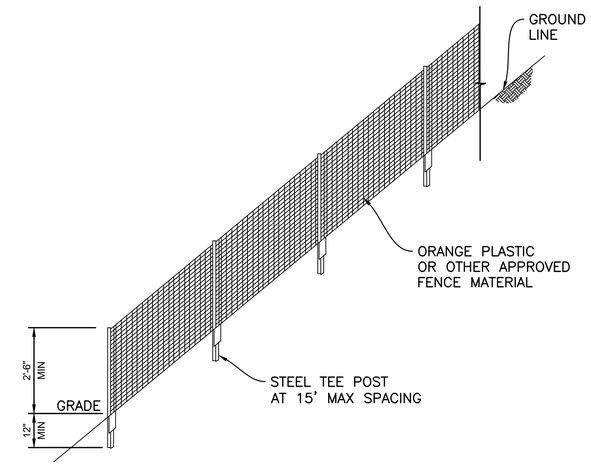
"I CERTIFY UNDER PENALTY OF LAW THAT THIS DOCUMENT AND ALL ATTACHMENTS WERE PREPARED UNDER MY DIRECTION AND SUPERVISION IN ACCORDANCE WITH A SYSTEM DESIGNED TO ASSURE THAT QUALIFIED PERSONNEL PROPERLY GATHERED AND EVALUATED THE INFORMATION SUBMITTED. BASED ON MY INQUIRE OF THE OPERATOR OR OPERATORS WHO MANAGE THE SYSTEM, OR THOSE OPERATORS DIRECTLY RESPONSIBLE FOR GATHERING INFORMATION, THE INFORMATION SUBMITTED IS, TO THE BEST OF MY KNOWLEDGE AND BELIEF, TRUE, ACCURATE, AND COMPLETE. I AM AWARE THAT THERE ARE SIGNIFICANT PENALTIES FOR SUBMITTING FALSE INFORMATION, INCLUDING THE POSSIBILITY OF FINE AND IMPRISONMENT FOR KNOWING VIOLATIONS."

	CONSTRUCTION ACTIVITIES AND SCHEDULING	
	ACTIVITY	SCHEDULE
	INSTALL ALL BMPs NEEDED AND ASSOCIATED WITH THE GRADING PHASE SUCH AS STABILIZED CONSTRUCTION ENTRANCES, SILT FENCE, DIVERSIONS AND TERRACES.	PRIOR TO ANY STRIPPING OF EXISTING VEGETATION OR GRADING.
	PROCEED WITH STRIPPING OF EXISTING VEGETATION AND GRADING PLAN, WHILE DISTURBING NO MORE THAN NECESSARY.	AFTER INSTALLING ALL BMPs NEEDED AND ASSOCIATED WITH THE GRADING PHASE. FURTHERMORE, OBTAIN APPROVAL FROM THE NPS BEFORE THE START OF ANY STRIPPING OF EXISTING VEGETATION OR GRADING.
	PROCEED WITH INFRASTRUCTURE INSTALLATION.	INFRASTRUCTURE INSTALLATION MUST OCCUR PRIOR TO ANY LOT DEVELOPMENT.
	IMPLEMENT THE INSTALLATION OF TEMPORARY SEEDING, PERMANENT SEEDING, AND/OR MULCHING.	STABILIZATION MEASURES MUST BE INITIATED AS SOON AS POSSIBLE IN PORTIONS OF THE SITE WHERE CONSTRUCTION ACTIVITIES HAVE TEMPORARILY OR PERMANENTLY CEASED, BUT IN NO CASE MORE THAN 14 DAYS AFTER THE CONSTRUCTION ACTIVITY IN THAT PORTION OF THE SITE HAS TEMPORARILY OR PERMANENTLY CEASED.
	IMPLEMENT THE INSTALLATION OF ALL BMPs NEEDED AND ASSOCIATED WITH THE BUILDING PHASE.	BUILDING PHASE BMPs MUST BE INSTALLED CONCURRENTLY WITH LOT DEVELOPMENT.
	PROCEED WITH REMOVAL OF BMPs.	BMPs MAY NOT BE REMOVED UNTIL EACH IMPACTED DRAINAGE BASIN HAS BEEN FULLY DEVELOPED. FULL DEVELOPMENT SHALL MEAN INSTALLATION OF PAVEMENT, BUILDINGS, AND UTILITIES, LANDSCAPING, AND FULLY ESTABLISHED PERMANENT SEEDING. FURTHERMORE, CONTRACTING OFFICER APPROVAL MUST BE OBTAINED BEFORE THE REMOVAL OF ANY BMPs.

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — Date Signed 5/1/2026 — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: G. GIORLANDO TECH. REVIEW: J. GLOVER DATE: MAY 2026	SUB SHEET NO. 002 C508	TITLE OF SHEET GENERAL EROSION AND SEDIMENT CONTROL NOTES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 30 OF 99

4/17/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\GEN\002-C508.dwg



CF — CF — 1 / 002C509 CONSTRUCTION FENCE DETAIL

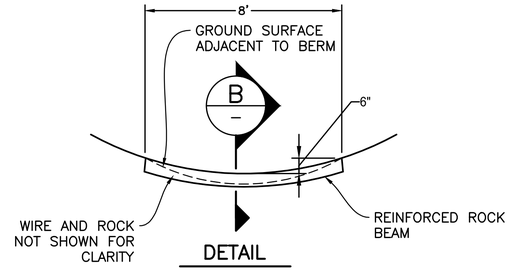
NOT TO SCALE

CONSTRUCTION FENCE INSTALLATION NOTES

1. SEE PLAN VIEW FOR:
   - LOCATION AND LENGTH OF FENCE.
2. CONSTRUCTION FENCE INDICATED ON INITIAL GESC PLAN SHALL BE INSTALLED PRIOR TO OTHER BMPS AND ANY LAND-DISTURBING ACTIVITIES.
3. STEEL TEE POSTS SHALL BE UTILIZED FOR SUPPORT OF CONSTRUCTION FENCE. MAXIMUM SPACING FOR TEE POSTS SHALL BE 15'.

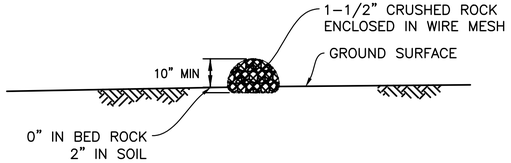
CONSTRUCTION FENCE MAINTENANCE NOTES

1. ANY DAMAGED FENCE SHALL BE REPAIRED ON A DAILY BASIS.
2. FENCE SHALL BE REMOVED AT THE END OF CONSTRUCTION. IF ANY DISTURBED AREA EXISTS AFTER FENCE REMOVAL, IT SHALL BE DRILL SEEDED AND CRIMP MULCHED OR OTHERWISE STABILIZED IN A MANNER APPROVED BY THE GOVERNING AGENCY.



DETAIL

NOT TO SCALE



B – SECTION

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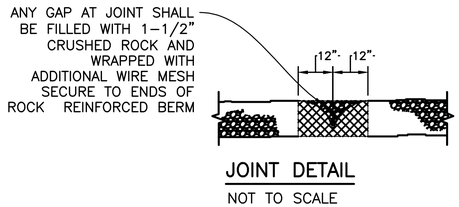

JOINT DETAIL

NOT TO SCALE

REINFORCED ROCK BERM INSTALLATION NOTES

1. SEE PLAN VIEW FOR:
   - LOCATIONS OF REINFORCED ROCK BERMS.
2. REINFORCED ROCK BERM SECTION APPLIES TO CULVERT INLET FILTER AND INLET PROTECTION.
3. CRUSHED ROCK SHALL BE FRACTURED FACE (ALL SIDES) AND SHALL COMPLY WITH GRADATION SHOWN ON GESC DETAIL SHEET (1-1/2" MINUS). RECYCLED CONCRETE MEETING THIS GRADATION MAY BE USED.
4. WIRE MESH SHALL BE FABRICATED OF 10 GAUGE WIRE TWISTED INTO A MESH WITH A MAXIMUM OPENING OF 1.0 INCH (COMMONLY TERMED "CHICKEN WIRE"). ROLL WIDTH SHALL BE 48-INCHES.
5. WIRE MESH SHALL BE SECURED USING "HOG RINGS" OR WIRE TIES AT 6-INCH CENTERS ALONG ALL JOINTS AND AT 2-INCH CENTERS ON ENDS OF BERM.
6. FOR CONCENTRATED FLOW AREAS THE ENDS OF THE REINFORCED ROCK BERM SHALL BE 12" HIGHER THAN THE CENTER OF THE BERM.

REINFORCED ROCK BERM MAINTENANCE NOTES

1. THE GESC MANAGER SHALL INSPECT ROCK BERMS WEEKLY, DURING AND AFTER ANY STORM EVENT AND MAKE REPAIRS OR CLEAN OUT AS NECESSARY.
2. SEDIMENT ACCUMULATED UPSTREAM OF REINFORCED ROCK BERM SHALL BE REMOVED WHEN THE SEDIMENT DEPTH UPSTREAM OF FILTER IS WITHIN 5 INCHES OF THE CREST.
3. REINFORCED ROCK BERMS ARE TO REMAIN IN PLACE UNTIL THE UPSTREAM DISTURBED AREA IS STABILIZED AND GRASS COVER IS APPROVED.
4. WHEN REINFORCED ROCK BERMS ARE REMOVED, ANY DISTURBED AREA SHALL BE DRILL SEEDED AND CRIMP MULCHED OR OTHERWISE STABILIZED IN A MANNER APPROVED BY THE OWNER.
5. FOR MORE INFORMATION SEE SPECIFICATION 31 25 00 SOIL EROSION AND SEDIMENT CONTROL.

RRB — 2 / 002C509 REINFORCED ROCK BERM

NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — Date Signed 5/1/26 — ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	G. YOUNG	002 C509	GENERAL CIVIL DETAILS – GESC CONTINUED	113 182752
	CADD C. WITTENMEIER			PMIS/PKG NO. 318719
	TECH. REVIEW: J. GLOVER		NORTH RIM - GRAND CANYON NATIONAL PARK	SHEET 31 OF 99
	DATE: MAY 2026			

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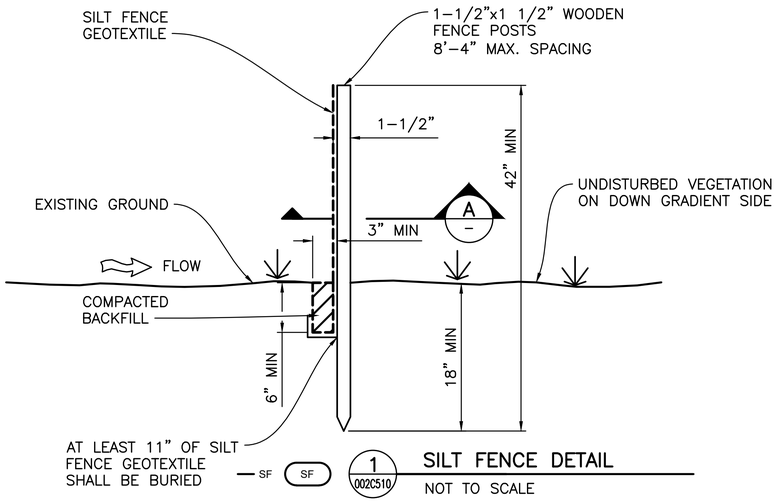
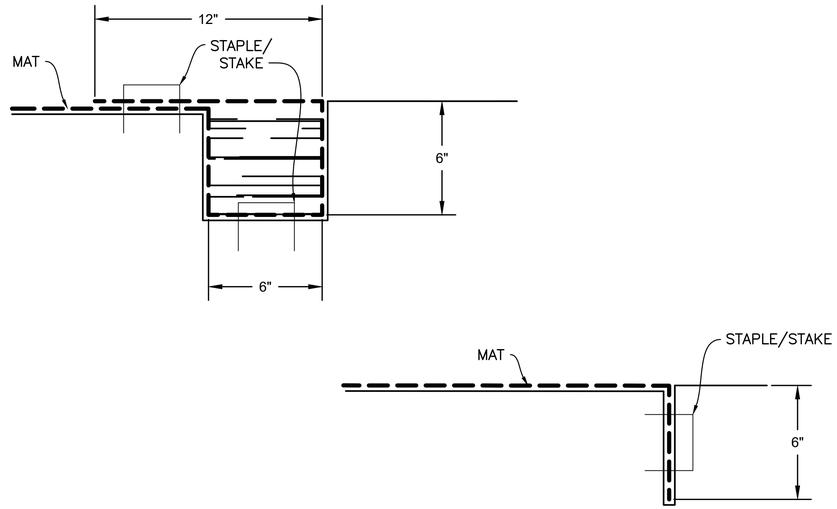


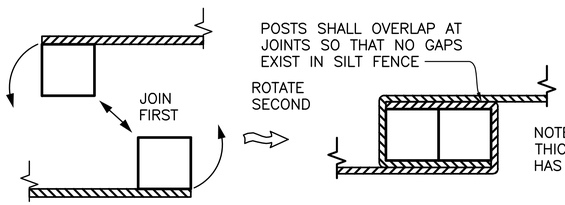
1 002C510 **SILT FENCE DETAIL**

NOT TO SCALE



2 002C510 **EROSION CONTROL BLANKET**

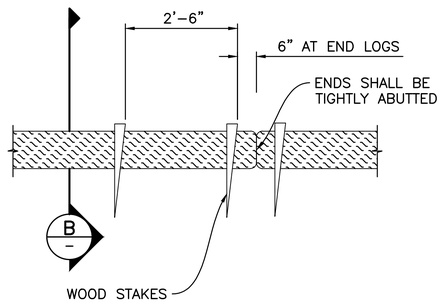
NOT TO SCALE



POST SHALL BE JOINED AS SHOWN, THEN ROTATED 180° IN DIRECTION SHOWN AND DRIVEN INTO THE GROUND

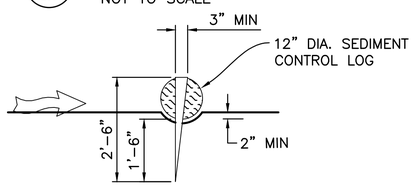
A – **JOINTS – SECTION**

NOT TO SCALE



SCL 3 002C510 **SEDIMENT CONTROL LOG**

NOT TO SCALE



B – **SECTION**

NOT TO SCALE

SEDIMENT CONTROL LOG INSTALLATION NOTES

1. SEE PLAN VIEW FOR: – LOCATION AND LENGTH OF SEDIMENT CONTROL LOG.
2. SEDIMENT CONTROL LOGS INDICATED ON INITIAL GESC PLANS SHALL BE INSTALLED PRIOR TO ANY LAND–DISTURBING ACTIVITY.
3. SEDIMENT CONTROL LOGS SHALL CONSIST OF STRAW, COMPOST, EXCELSIOR, OR COCONUT FIBER.
4. NOT FOR USE IN CONCENTRATED FLOW AREAS.
5. THE SEDIMENT CONTROL LOG SHALL BE TRENCHED INTO THE GROUND A MINIMUM OF 2".

SEDIMENT CONTROL LOG MAINTENANCE NOTES

1. THE GESC MANAGER SHALL INSPECT SEDIMENT CONTROL LOGS DAILY, DURING AND AFTER ANY STORM EVENT AND MAKE REPAIRS OR CLEAN OUT UPSTREAM SEDIMENT AS NECESSARY.
2. SEDIMENT ACCUMULATED UPSTREAM OF SEDIMENT CONTROL LOGS SHALL BE REMOVED WHEN THE UPSTREAM SEDIMENT DEPTH IS WITHIN 1/2 THE HEIGHT OF THE CREST OF LOGS.
3. SEDIMENT CONTROL LOG SHALL BE REMOVED AT THE END OF CONSTRUCTION. IF ANY DISTURBED AREA EXISTS AFTER REMOVAL, IT SHALL BE DRILL SEEDED AND CRIMP MULCHED OR OTHERWISE STABILIZED IN A MANNER APPROVED BY THE GOVERNING AGENCY.

SILT FENCE INSTALLATION NOTES:

1. SEE PLAN VIEW FOR: – LOCATION AND LENGTH OF FENCE.
2. ANCHOR TRENCH SHALL BE EXCAVATED WITH TRENCHER, OR WITH SILT FENCE INSTALLATION MACHINE; NO ROAD GRADERS, BACKHOES, ETC. SHALL BE USED. TRENCH SHALL BE COMPACTED BY HAND; WITH "JUMPING JACK", OR BY WHEEL ROLLING. COMPACTION SHALL BE SUCH THAT SILT FENCE RESISTS BEING PULLED OUT OF ANCHOR TRENCH BY HAND.
3. SILT FENCE GEOTEXTILE SHALL MEET THE FOLLOWING REQUIREMENTS:
   – 6 TO 12 GALLONS PER MINUTE PER SQUARE FOOT FLOW CAPACITY.
   – 90 LB. TENSILE STRENGTH PER ASTM D1622.
   – UV DESIGN AT 500 HRS. MIN. 70% STRENGTH RETAINED PER ASTM D4355.
4. SILT FENCE SHALL BE INSTALLED PRIOR TO ANY LAND–DISTURBING ACTIVITIES.

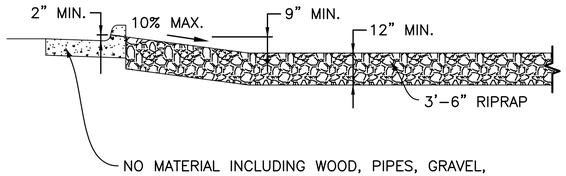
SILT FENCE MAINTENANCE NOTES:

1. THE GESC MANAGER SHALL INSPECT SILT FENCE DAILY DURING AND AFTER AND STORM EVENT AND MAKE REPAIRS OR CLEAN OUT UPSTREAM SEDIMENT AS NECESSARY.
2. SEDIMENT ACCUMULATED UP STREAM OF SILT FENCE SHALL BE REMOVED WHEN THE UPSTREAM SEDIMENT REACHES A DEPTH OF 6 INCHES.
3. SILT FENCE SHALL BE REMOVED WHEN THE UPSTREAM DISTURBED AREA IS STABILIZED AND GRASS COVER IS APPROVED BY THE TOWN. IF ANY DISTURBED AREA EXISTS AFTER REMOVAL, IT SHALL BE SEEDED AND MULCHED OR OTHERWISE STABILIZED IN A MANNER APPROVED BY THE OWNER.

REGISTERED PROFESSIONAL ENGINEER – CERTIFICATE NO. 75262 – GREGORY YOUNG – ARIZONA, U.S.A.

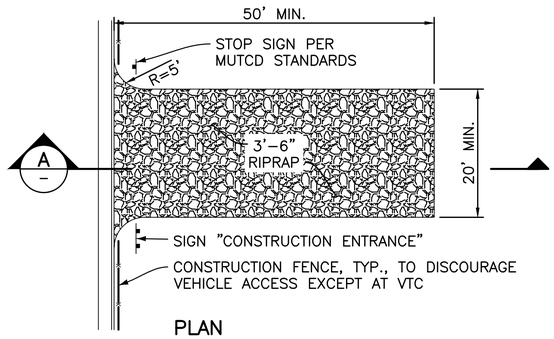
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: C. WITTENMEIER TECH. REVIEW: J. GLOVER DATE: MAY 2026	SUB SHEET NO. 002 C510	TITLE OF SHEET GENERAL CIVIL DETAILS – GESC CONTINUED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 32 OF 99

4/17/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\GEN\002-C501.dwg



A/– SECTION

SCALE: 1/2" = 1'–0"

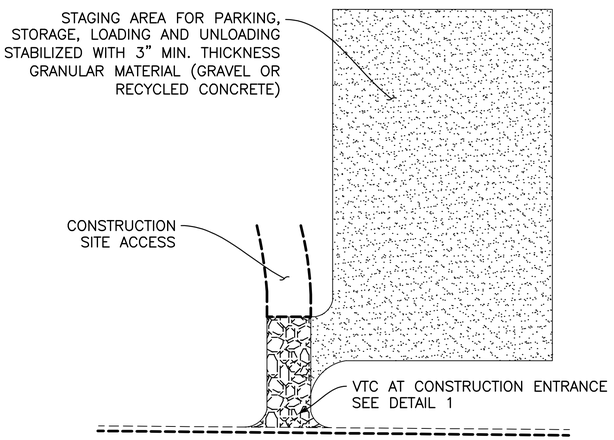


PLAN

SCALE: 1" = 10'–0"

VTC 1/002C511 VEHICLE TRACKING CONTROL

NOT TO SCALE



PLAN

SCALE: 1" = 40'–0"

STABILIZED STAGING AREA INSTALLATION NOTES

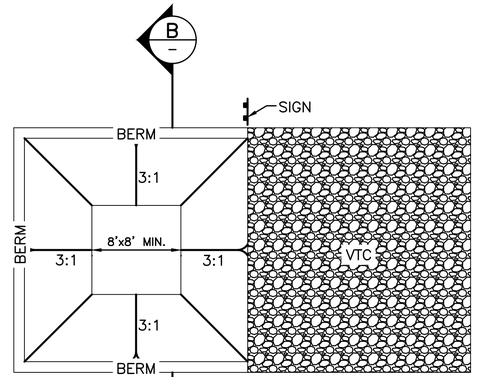
1. SEE PLAN VIEW FOR GENERAL LOCATION OF STAGING AREA. CONTRACTOR MAY MODIFY LOCATION AND SIZE OF STABILIZED STAGING AREA WITH TOWN APPROVAL.
2. STABILIZED STAGING AREA SHALL BE LARGE ENOUGH TO FULLY CONTAIN PARKING, STORAGE, AND UNLOADING AND LOADING OPERATIONS.
3. IF REQUIRED BY THE TOWN, SITE ACCESS ROADS SHALL BE STABILIZED IN THE SAME MANNER AS THE STAGING AREA.
4. STAGING AREA SHALL BE STABILIZED PRIOR TO ANY OTHER OPERATIONS ON THE SITE.
5. THE STABILIZED STAGING AREA SHALL CONSIST OF A MINIMUM OF 3" OF GRANULAR MATERIAL (GRAVEL OR RECYCLED CONCRETE).

STABILIZED STAGING AREA MAINTENANCE NOTES

1. THE GESC MANAGER SHALL INSPECT THE STABILIZED STAGING AREA WEEKLY, DURING AND AFTER ANY STORM EVENT AND MAKE REPAIRS OR CLEAN OUT UPSTREAM SEDIMENT AS NECESSARY.
2. GESC MANAGER SHALL PROVIDE ADDITIONAL THICKNESS OF GRANULAR MATERIAL IF ANY RUTTING OCCURS OR UNDERLYING SUBGRADE BECOMES EXPOSED.
3. STABILIZED STAGING AREA SHALL BE ENLARGED IF NECESSARY TO CONTAIN PARKING, STORAGE, AND UNLOADING AND LOADING OPERATIONS.
4. ANY ACCUMULATED DIRT OR MUD SHALL BE REMOVED FROM THE SURFACE OF THE STABILIZED STAGING AREA.
5. THE STABILIZED STAGING AREA SHALL BE REMOVED AT THE END OF CONSTRUCTION. THE GRANULAR MATERIAL SHALL BE REMOVED OR, IF APPROVED BY THE TOWN, USED ON SITE, AND THE AREA TOPSOILED, DRILL SEEDED AND CRIMP MULCHED OR OTHERWISE STABILIZED.

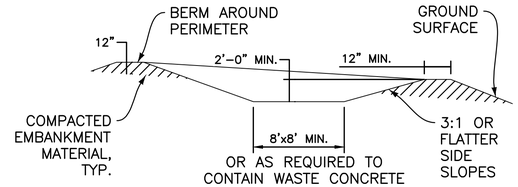
SSA 3/002C511 STABILIZED STAGING AREA

NOT TO SCALE



PLAN

NO SCALE



B/– SECTION

NO SCALE

CWA

CONCRETE WASHOUT AREA INSTALLATION NOTES

1. SEE PLAN VIEW FOR–

   – LOCATIONS OF CONCRETE WASHOUT AREA
2. THE CONCRETE WASHOUT AREA SHALL BE INSTALLED PRIOR TO ANY CONCRETE PLACEMENT ON SITE.
3. VEHICLE TRACKING CONTROL IS REQUIRED AT THE ACCESS POINT.
4. SIGNS SHALL BE PLACED AT CONSTRUCTION ENTRANCE, AT THE WASHOUT AREA, AND ELSEWHERE AS NECESSARY TO CLEARLY INDICATE THE LOCATION OF THE CONCRETE WASHOUT AREA TO OPERATORS OF CONCRETE TRUCKS AND PUMP RIGS.
5. EXCAVATED MATERIAL SHALL BE UTILIZED IN PERIMETER BERM CONSTRUCTION.

CONCRETE WASHOUT AREA MAINTENANCE NOTES

1. THE CONCRETE WASHOUT AREA SHALL BE REPAIRED AND ENLARGED OR CLEANED OUT AS NECESSARY TO MAINTAIN CAPACITY FOR WASTED CONCRETE.
2. AT THE END OF CONSTRUCTION, ALL CONCRETE SHALL BE REMOVED FROM SITE AND DISPOSED OF AT AN APPROVED WASTE SITE.
3. WHEN THE CONCRETE WASHOUT AREA IS REMOVED, THE DISTURBED AREA SHALL BE DRILL SEEDED AND CRIMP MULCHED OR OTHERWISE STABILIZED IN A MANNER APPROVED BY THE GOVERNING AGENCY.
4. INSPECT WEEKLY, DURING AND AFTER ANY STORM EVENT.

2/002C511 CONCRETE WASH OUT DETAIL

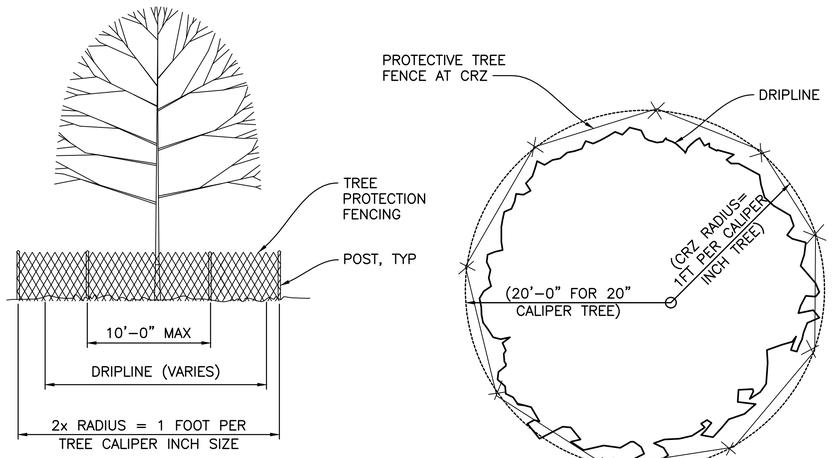
NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER – CERTIFICATE NO. 75262 – GREGORY YOUNG – Date Signed 5/1/2026 – ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG ADD C. WITTENMEIER TECH. REVIEW: J. GLOVER DATE: MAY 2026	SUB SHEET NO. 002 C511	TITLE OF SHEET GENERAL CIVIL DETAILS – GESC CONTINUED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 33 OF 99

4/17/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\GEN\002-C501.dwg

NOTES:
1. TREE PROTECTION TO BE INSTALLED PRIOR TO CONSTRUCTION ACTIVITY.
2. PROTECTIVE TREE FENCE SHALL BE PROVIDED AT CRITICAL ROOT ZONE (CRZ). ANY TREES THAT WILL NOT MEET THE TREE PRESERVATION FENCE REQUIREMENT AT CRZ MUST BE SAVED PER THE CONTRACTORS DISCRETION.
3. ALL ROOT PRUNING AND TREE CARE DURING CONSTRUCTION SHALL BE MONITORED AND ADMINISTERED BY A CERTIFIED ARBORIST.
4. THE RADIUS OF THE CRZ = 12 x TREE CALIPER INCHES. MEASUREMENT OF TREE CALIPER INCH SHALL MEET EITHER THE AMERICAN STANDARD FOR NURSERY STOCK OR THE AMERICAN FORESTS DEFINITION.

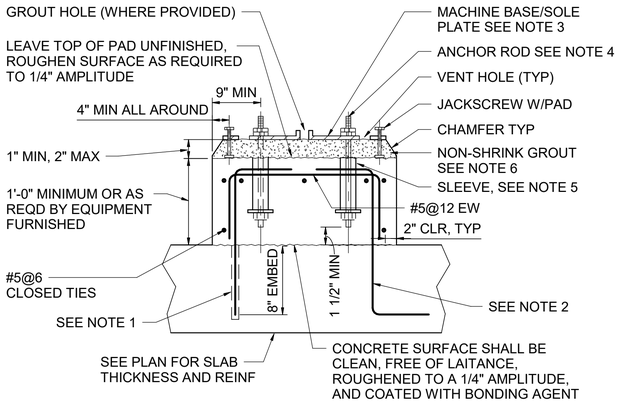


1 / 002C512 TREE PROTECTION
NOT TO SCALE

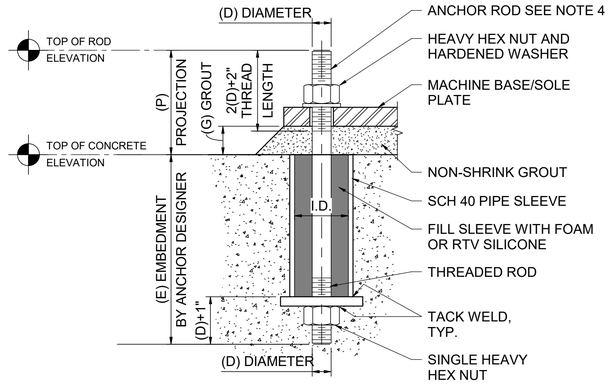
4/17/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\GEN\002-C501.dwg

REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 75282 GREGORY YOUNG Date Signed 5/11/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD C. WITTENMEIER TECH. REVIEW: J. GLOVER DATE: MAY 2026	SUB SHEET NO. 002 C512	TITLE OF SHEET GENERAL CIVIL DETAILS – GESC CONTINUED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 34 OF 99



CONCRETE EQUIPMENT PAD



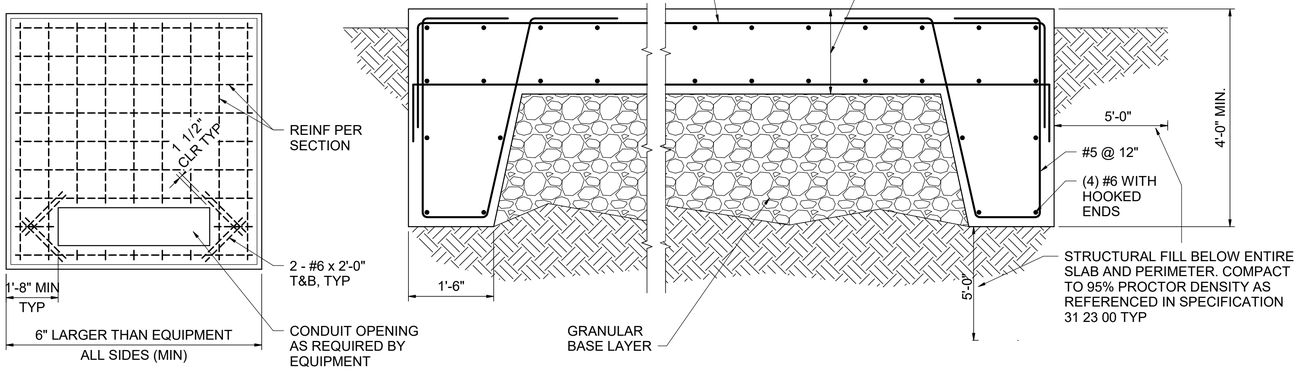
POST TENSIONED ANCHOR ROD

NOTES:

1. FOR CONCRETE EQUIPMENT SUPPORT BASES ON EXSITING SLABS, DRILL INTO SLAB TO INDICATED DEPTH AT 12" ON CENTER AROUND PERIMETER OF CONCRETE EQUIPMENT SUPPORT BASE AND SET #5 DOWEL HOOKED AS SHOWN USING ADHESIVE FOR CONCRETE ANCHORS. THICKNESS TO BE COORDINATED WITH EQUIPMENT SUPPLIER. THICKEN SLAB ON GRADE BELOW EQUIPMENT PAD AS REQUIRED TO MAINTAIN MINIMUM 3" COVER ON EQUIPMENT ANCHOR BOLTS.
2. FOR CONCRETE EQUIPMENT SUPPORT BASES ON NEW SLABS, PROVIDE #5 DOWELS HAVING TWO HOOKED ENDS AT 12" ON CENTER AROUND PERIMETER, MINIMUM.
3. MACHINE BASE/SOLE PLATE DIMENSION SHALL BE AS REQUIRED BY THE APPROVED SHOP DRAWINGS.
4. ANCHOR ROD AS RECOMMENDED BY THE EQUIPMENT MANUFACTURER. RODS SHALL BE HELD IN POSITION WITH A TEMPLATE WHILE EQUIPMENT SUPPORT BASE IS BEING CAST IN PLACE. TAPE OR GREASE ALL PORTIONS OF THE ANCHOR ROD THAT EXTENDS ABOVE THE TOP OF CONCRETE TO THE TOP OF THE BASE PLATE SO THAT THE GROUT WILL NOT BOND TO THE ANCHOR ROD.
5. SLEEVES SHALL BE STEEL PIPES AS SHOWN IN POST TENSIONED ANCHOR ROD DETAIL.
6. GROUT VENT HOLES SHALL BE INSTALLED IN ALL EQUIPMENT BASE PLATES THAT ARE LARGER THAN 12" WIDE SO THAT GROUT IS ABLE TO COMPLETELY FILL IN UNDER THE ENTIRE BASE PLATE.
7. ONCE NON-SHRINK GROUT HAS REACHED DESIGN STRENGTH, BACK OFF JACK SCREWS AND POST TENSION ANCHORS TO REQUIRED TENSION LOAD SPECIFIED BY ANCHOR DESIGNER. ANCHOR BOLTS SHALL BE PRETENSIONED BY CONTRACTOR'S ENGINEER (ANCHOR DESIGNER).
8. FOLLOWING CONSTRUCTION, CARE SHOULD BE TAKEN TO ENSURE THAT REQUIRED TENSION IS MAINTAINED WITHIN THE ANCHOR FOR FULL DURATION OF EQUIPMENT LIFESPAN. THIS SHALL INCLUDE ANNUAL TENSION CHECKS OF THE ANCHORS FOR THE FIRST 5 YEARS OF OPERATION, FOLLOWED BY TENSION CHECKS EVERY 3 YEARS.
9. FOR EQUIPMENT 50 HP AND BELOW, ANCHOR SLEEVES MAY BE OMITTED.
10. WHERE EQUIPMENT SKIDS ARE PROVIDED FOR EQUIPMENT MOUNTING, SKIDS SHALL BE FILLED WITH GROUT AFTER INSTALLATION , UNLESS NOTED OTHERWISE BY THE MANUFACTURER.
11. EQUIPMENT PAD SHOWN APPLIES FOR ROTATING EQUIPMENT WHERE SPECIFICALLY REFERENCED ON THE DRAWINGS.

1 003S501 CONCRETE PAD FOR DYNAMIC EQUIPMENT

NOT TO SCALE

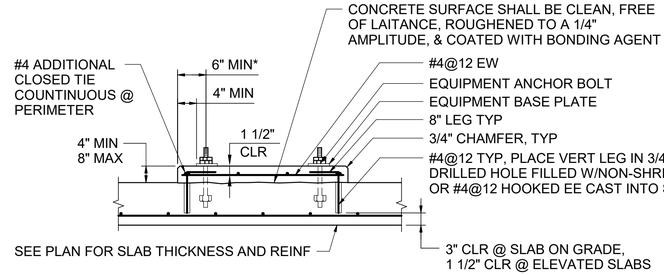


PLAN: CONDUIT OPENING (WHERE OCCURS)

SECTION

3 003S501 SITE PAD DETAIL

NOT TO SCALE

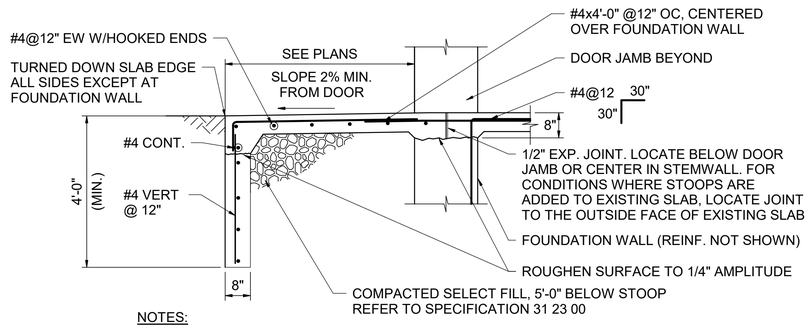


NOTES:

1. PROVIDE ABOVE PAD UNDER ALL ELECTRICAL AND MECHANICAL EQUIPMENT SUPPORTED ON STRUCTURAL SLABS UNLESS NOTED OTHERWISE. ALSO PROVIDE FOR EQUIPMENT WEIGHING LESS THAN 5000 POUNDS WHICH ARE SUPPORTED ON GRADE, OR WHERE SPECIFICALLY NOTED ON PLANS. FOR PAD HEIGHT GREATER THAN 8", SEE TALL EQUIPMENT SUPPORT PAD.
2. PAD THICKNESS SHALL BE A MINIMUM OF 4". CONTRACTOR SHALL VERIFY THE PAD THICKNESS WITH EQUIPMENT SUPPLIER. THICKEN SLAB ON GRADE BELOW EQUIPMENT PAD AS REQUIRED TO MAINTAIN MINIMUM 3" COVER ON EQUIPMENT ANCHOR BOLTS.
3. PROVIDE EMBEDDED CHANNELS FOR ANCHORING ELEC EQUIPMENT WHERE REQ'D.
4. BEFORE EQUIPMENT SUPPORT PADS ARE CAST, THE PAD SIZES AND REINFORCING SHALL BE APPROVED BY THE CONTRACTING OFFICER AS BEING CAPABLE OF SUPPORTING EQUIPMENT TO BE PLACED THEREON. EQUIPMENT BASE DIMENSIONS SHALL BE THE LARGER AS DETERMINED BY THE EQUIPMENT MANUFACTURER OR AS INDICATED ON THE DRAWINGS. SUBMIT ALL EQUIPMENT DIMENSIONS AND LOADS TO CONTRACTING OFFICER. THE SIZE, NUMBER, TYPE, LOCATION, AND THREAD PROJECTION OF THE ANCHOR BOLTS (AB) SHALL BE AS DETERMINED BY THE EQUIPMENT MANUFACTURER AND SHALL BE AS APPROVED BY THE CONTRACTING OFFICER. AB SHALL BE HELD IN POSITION WITH A TEMPLATE WHILE EQUIPMENT PAD IS CAST.
5. 6" MINIMUM PAD EDGE DIMENSION TO EQUIPMENT AB APPLIES FOR ALL NON-ELECTRICAL EQUIPMENT SUPPORT PADS. AT ELECTRICAL EQUIPMENT PROVIDE 1" FROM EDGE OF EQUIPMENT TO EDGE OF EQUIPMENT PAD.
6. THE SIZE, NUMBER, TYPE, LOCATION, AND THREAD PROJECTION OF THE ANCHOR BOLTS SHALL BE AS DETERMINED BY THE EQUIPMENT MANUFACTURER AND SHALL BE AS APPROVED BY THE CONTRACTING OFFICER. ANCHOR BOLTS SHALL BE HELD IN POSITION WITH A TEMPLATE WHILE EQUIPMENT BASE IS BEING CAST, SEE DETAIL 2/003S502.
7. EQUIPMENT BASES SHALL BE INSTALLED LEVEL UNLESS SPECIFIED OTHERWISE, TOLERANCE IS 1/8" ACROSS PLAN DIAGONALS.

2 003S501 TYPICAL SHALLOW EQUIPMENT SUPPORT PAD

NOT TO SCALE



NOTES:

1. AT STAIR STOOPS, PROVIDE SIMILAR STOOP WITH TURNED-DOWN EDGE.
2. TURNED-DOWN EDGE MAY BE OMITTED AT EDGES WHERE SIDEWALK CONTINUES BEYOND.
3. PROVIDE 1/2" EXPANSION JOINT PERPENDICULAR TO BUILDING WHERE STOOPS INTERFACE ADJACENT FINISHES.

4 003S501 STOOP DETAIL

NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed 02/11/2026 ARIZONA, U.S.A.

REGISTERED PROFESSIONAL ENGINEER (Civil) CERTIFICATE NO. 85509 JOHN F. CORONADO Date Signed 02/11/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. CORONADO DRAWN: R. NELSON TECH. REVIEW: M. HERRERA DATE: MAY 2026	SUB SHEET NO. 003 S501	TITLE OF SHEET GENERAL STRUCTURAL DETAILS - PADS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 35 OF 99

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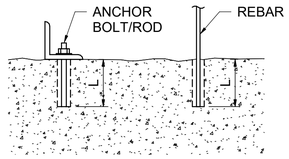
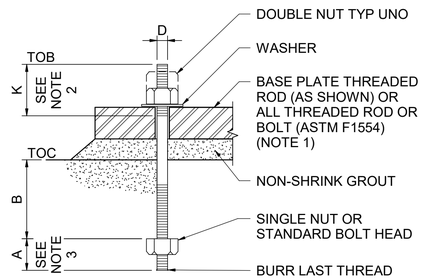


NOTES:

1. EPOXY SHALL BE PER SPECIFICATIONS.
2. EMBEDMENT LENGTHS SHOWN ARE MINIMUM. FOLLOW MANUFACTURER'S RECOMMENDATIONS FOR INSTALLATION.
3. DO NOT USE ADHESIVE ANCHORS FOR UPWARDLY INCLINED APPLICATIONS.
4. EMBED LENGTHS SHOWN REFLECT CRACKED CONCRETE, SEISMIC LOADING CONDITIONS USING EPOXY ADHESIVE PER PROJECT SPECIFICATIONS. FOR ADHESIVE WITH A LOWER BOND STRENGTH THAN THOSE LISTED IN THE SPECIFICATIONS, CONSULT CONTRACTING OFFICER FOR ALTERNATE EMBEDMENT.
5. FOR ANCHORS WITH 1.5L OF EDGE, CONSULT CONTRACTING OFFICER.

ADHESIVE ANCHOR SCHEDULE			
REINFORCING BARS		ANCHOR BOLTS/RODS	
BAR SIZE	EMBEDMENT LENGTH (L)	DIA (IN)	EMBEDMENT LENGTH (L)
#3	4"	3/8"	5"
#4	5"	1/2"	6"
#5	6"	5/8"	7"
#6	7"	3/4"	8"
#7	8"	7/8"	9"
#8	9"	1"	10"

1 003S502 ADHESIVE ANCHOR
NOT TO SCALE

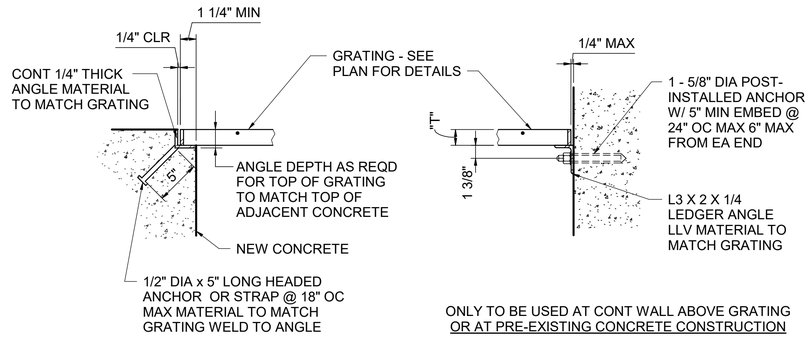


NOTES:

1. PROVIDE SST ANCHOR BOLTS WHERE INDICATED IN SECTIONS AND DETAILS. REFER TO SPECIFICATIONS FOR ANCHOR BOLT FINISHES.
2. STANDARD BOLT THREAD LENGTH MAY BE USED WHERE APPLICABLE.
3. DIMENSION IN SCHEDULE OR STANDARD BOLT HEAD.

SCHEDULE-ANCHOR BOLT TYPE A			
D	A	B	K
3/8"	1"	6"	2 3/4"
1/2"	1 1/4"	8"	3"
5/8"	1 1/2"	10"	3 1/4"
3/4"	1 3/4"	12"	3 1/2"
7/8"	2"	14"	3 3/4"
1"	2 1/4"	16"	4"
1 1/8"	2 1/2"	18"	4 1/4"
1 1/4"	2 3/4"	20"	4 1/2"
1 3/8"	3"	22"	4 3/4"
1 1/2"	3 1/4"	24"	5"
1 3/4"	3 3/4"	28"	5 1/2"
2"	4 1/4"	32"	6"
2 1/2"	5 1/2"	48"	7"
3"	6 1/4"	66"	8"

2 003S502 ANCHOR BOLT DETAIL
NOT TO SCALE

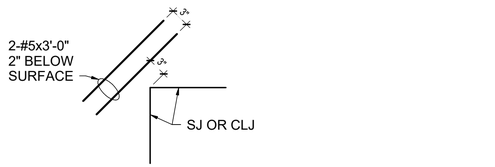


ONLY TO BE USED AT CONT WALL ABOVE GRATING
OR AT PRE-EXISTING CONCRETE CONSTRUCTION

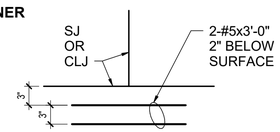
NOTES:

1. GRATING SIZE PER CONTRACT DOCUMENTS.
2. ALL ENDS AND OPENINGS SHALL BE BANDED, SEE SPECIFICATION.
3. ATTACH GRATING TO ALL SUPPORT ANGLES WITH BOLTED CLIPS, SPACED AT 2'-0" MAX CENTERS.
4. PROVIDE DISSIMILAR MATERIAL PROTECTION FOR ALUMINUM IN CONTACT WITH CONCRETE PER SPECIFICATION.

3 003S502 GRATING AND SUPPORT DETAIL
NOT TO SCALE



RE-ENTRANT CORNER



DISCONTINUOUS JOINT INTERSECTION

4 003S502 ADDITIONAL SLAB REINFORCING
NOT TO SCALE

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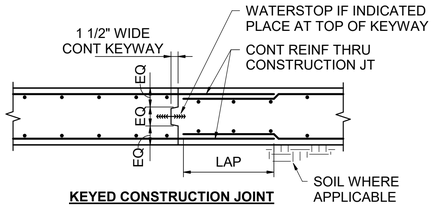
4/27/2026 11:23:17 AM

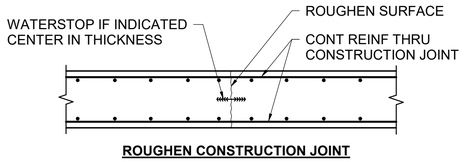
Registered Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed 4/27/2026 ARIZONA, U.S.A.

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 85509 JOSE CORONADO Date Signed 4/27/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. CORONADO CADD: R. NELSON TECH. REVIEW: M. HERRERA DATE: MAY 2026	SUB SHEET NO. 003 S502	TITLE OF SHEET GENERAL STRUCTURAL DETAILS - ANCHORING NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 36 OF 99



KEYED CONSTRUCTION JOINT



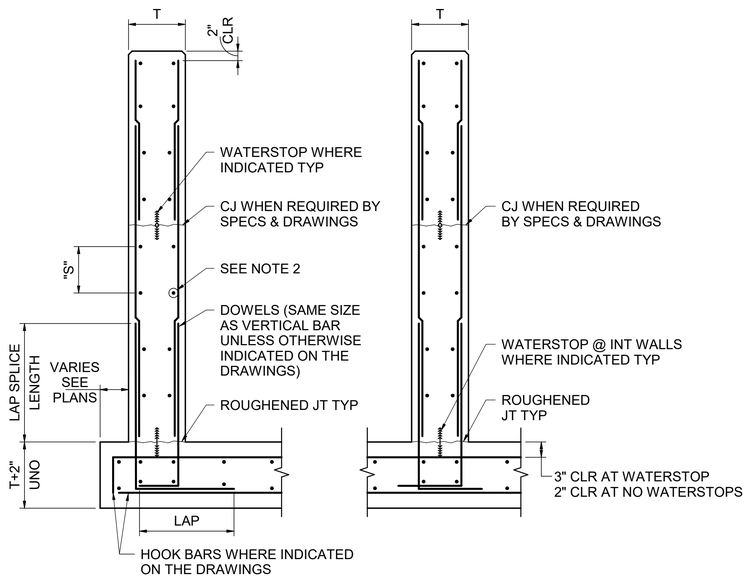
ROUGHEN CONSTRUCTION JOINT

NOTES:

1. SEE SPECIFICATION FOR REQUIREMENT TO TIE WATERSTOPS IN PLACE TO PREVENT MOVEMENT OR FOLDING OVER.
2. KEYED CONSTRUCTION JOINTS ARE REQUIRED FOR WALL CONSTRUCTION FOR WATER RETAINING STRUCTURES.

1 003S503 CONSTRUCTION JOINT

NOT TO SCALE

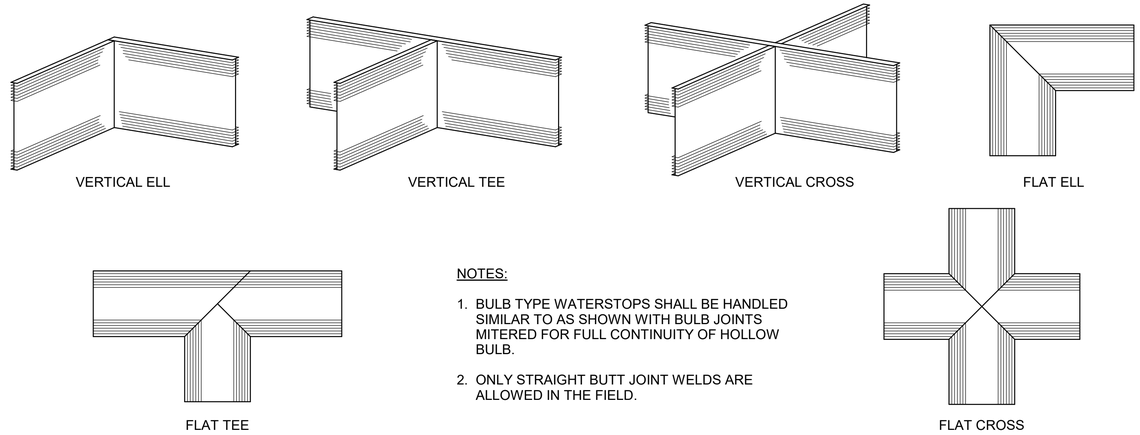


NOTES:

1. "S" = BAR SPACING INDICATED ON THE DRAWINGS.
2. SEE PLANS AND SECTIONS FOR HORIZONTAL REINF BAR LOCATION RELATIVE TO VERTICAL REINF BAR LOCATION (INSIDE OR OUTSIDE) SEE DRAWINGS (INSIDE SHOWN).

3 003S503 HORIZONTAL WALL JOINTS WITH WATERSTOPS

NOT TO SCALE



NOTES:

1. BULB TYPE WATERSTOPS SHALL BE HANDLED SIMILAR TO AS SHOWN WITH BULB JOINTS MITERED FOR FULL CONTINUITY OF HOLLOW BULB.
2. ONLY STRAIGHT BUTT JOINT WELDS ARE ALLOWED IN THE FIELD.

2 003S503 SHOP FABRICATED WATERSTOP

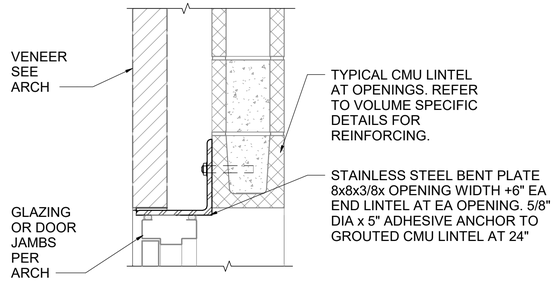
NOT TO SCALE

4 003S503 NOT USED

NOT TO SCALE

5 003S503 NOT USED

NOT TO SCALE



6 003S503 TYPICAL STEEL LINTEL AT OPENINGS

NOT TO SCALE

Registered Professional Engineer (Civil)
CERTIFICATE NO.
72480
COLIN MULDERICK
Date Signed 05/20/26
ARIZONA, U.S.A.

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO.
85509
JOSEPH R. CORONADO
Date Signed 05/20/26
ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	J. CORONADO	003 S503	GENERAL STRUCTURAL DETAILS - JOINTS	113 182752
	CADD R. NELSON			PMIS/PKG NO. 318719
	TECH. REVIEW: M. HERRERA			SHEET 37 OF 99
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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